
Software Specification

for

CareLog

Version 2.0 approved

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Organization: MediLink

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Revision History

Name	Date	Reason For Changes	Version
Lim Guo Zhan	27/8/2025	Collect related information	
Liaw Zi Shen	27/8/2025	Collect related information	
Leong Wen Kai	27/8/2025	Collect related information	
Lim Guo Zhan	10/9/2025	Finalize and add materials	

1. Introduction

1.1 Purpose

The purpose of the CareLog system is to create a unified, patient-centered platform that improves healthcare delivery by combining clinical records with personal, cultural, and emotional insights. For patients, the system provides tools to track daily well-being, log symptoms, manage preferences such as food and language, and book appointments. For medical staff, including doctors, nurses, and allied health professionals, CareLog streamlines daily tasks such as recording observations, monitoring vital signs, prescribing treatments, and coordinating care. It reduces errors, ensures continuity between shifts, and provides quick access to patient histories and preferences. For administrators, CareLog offers integrated management of users, schedules, resources, and billing, while generating accurate reports to support efficiency and compliance. For psychological counselors, CareLog ensures emotional health is included in care by supporting therapy notes, session scheduling, and secure communication. Overall, CareLog enhances collaboration, reduces inefficiencies, and improves patient experiences across the entire healthcare team.

1.2 Intended Audience and Reading Suggestions

This document is intended for multiple audiences involved in the design, development, and use of the CareLog system. Developers will use it to understand the functional and non-functional requirements, system features, and technical constraints necessary for implementation. Project managers will rely on it to monitor scope, track progress, and ensure that deliverables align with user needs and institutional goals. Testers will use the acceptance criteria and requirements to design test cases and validate system functionality. Hospital administrators and medical staff will reference the system features and stakeholder analyses to confirm that their needs are represented. Documentation writers and trainers will use it to create user manuals, training materials, and onboarding resources for patients and staff.

The document is organized into sections that cover the project purpose, scope, problem statements, stakeholder analyses, functional and non-functional requirements, system features, and acceptance criteria. Readers are advised to begin with the overview sections (Purpose and Scope) to gain context, followed by the stakeholder and problem analyses to understand user needs. Developers and testers should then focus on the requirements and system features, while administrators and trainers may prioritize the acceptance criteria and usability considerations.

2. Problem Statement

Doctor:

- Difficulty tracking patient histories, especially for patients with multiple illnesses or who have received multiple treatments
- Lack of real-time access to patient data, leading to delayed diagnosis and treatment decisions
- Inadequate communication tools lead to fragmented patient care and communication between colleagues.

Nurse:

- Difficult to track the patient's conditions and medications, especially for high-risk patients
- Lack of integrated communication systems with doctors and other medical staff, leading to coordination challenges.
- Inadequate tools to monitor and update patient vital signs in real-time.

Medical Staff:

- Inefficient patient record management leads to treatment delays and potential errors.
- Insufficient patient details influence decision-making and patient outcomes.
- Lack of real-time updates on patient status in treatment plans.

Psychological counsellors:

- Cannot check all the patient summaries or log to analyze their mental health.
- Difficulty in tracking the progress of mental health treatments due to incomplete or scattered records.
- Inability to communicate with nurses' results in the inability to collaborate effectively in patient treatment.

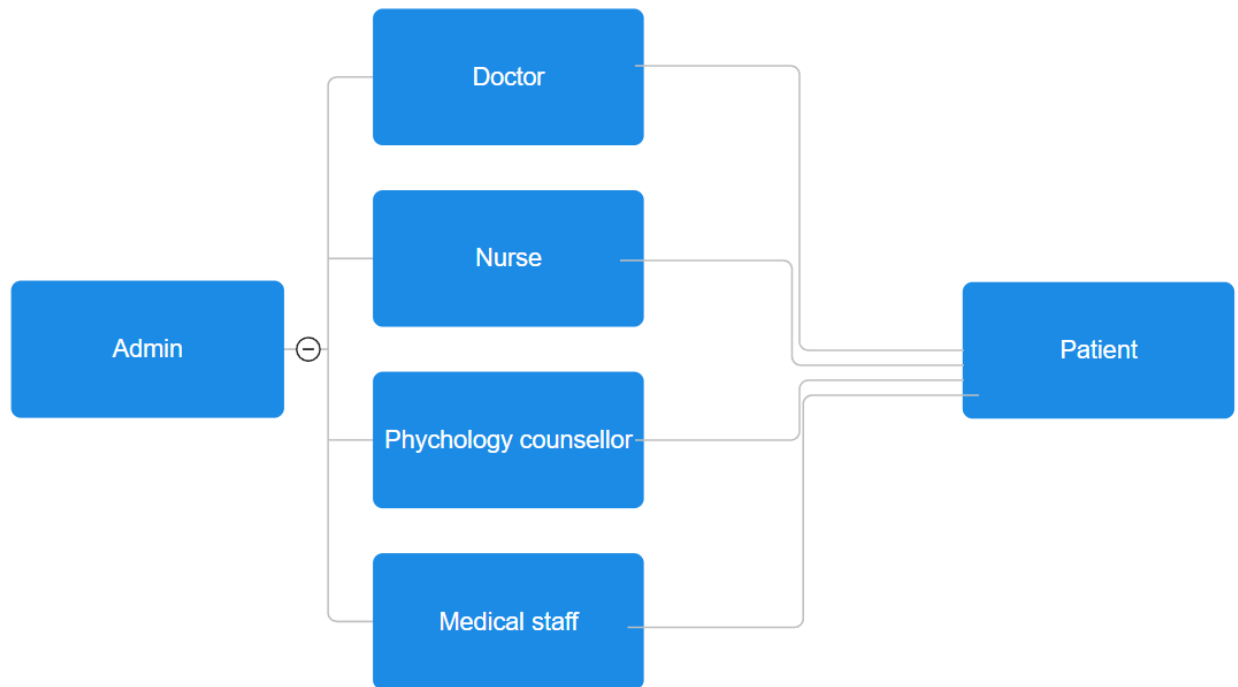
Patient

- Patients are unable to keep track of their daily well-being, medication schedules, and communication with healthcare staff
- Patients rely on in-person visits or calls to contact the staff. It can affect urgent concerns
- Patients forgot to take medication on time and struggle with recording symptoms such as pain levels and sleep quality
- The patient cannot express their preferences. For example, factors such as food, language, and the nurse's gender can lead to discomfort or satisfaction.
- Patient cannot hide or manage their health information themselves
- Appointment booking, reminders, health tracking and feedback are handled slowly and inefficiently

Admin

- A large number of hospitals continue to depend on manual work, paper documents, or incomplete software systems, which leads to operational inefficiencies.
 - Common problems include:
 - Patients, doctors, and employees losing track of files and documents.
 - Treatments getting postponed due to harm or risk.
 - Clients, patients, and doctors getting incorrect or incomplete information which leads to fragmented care.
 - A mix of weak or improper collaboration systems between different divisions of the same organization or different organizations.
 - The added administrative work and inefficiencies lower the overall patient experience.
 - Increased and insufficient healthcare providers make it increasingly difficult to optimally allocate and control the different healthcare resources.
 - Sensitive healthcare information is often vulnerable due to weak or improper collaboration systems.
 - There is no single integrated system through which healthcare admins can track and control the different doctors, patients, billing, and resources.
 - Healthcare service providers who continue to work without updates risk declining standards of care, team productivity, and relationship with patients.
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3. Stakeholder Analysis



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Hospital Admin

Goals

- Manage hospital operations efficiently.
- Minimize scheduling conflict/the overlaps in schedules (doctors, nurses, patients, and the facilities).
- Improve hospital efficiency and improve the patient experience.
- Maintain accurate medical, billing, and personal documents (staff records).
- Write clear reports for management and compliance.

Role

- Admin acts as the central controller of hospital operations.
- Admin assigns roles and permissions for doctors, nurses, counsellors and medical staff.
- Admin oversees patient flow from registration to discharge.
- Admin shall coordinate between the medical staff and the doctor or nurse.
- Admin manages billing, insurance, and compliance reporting.

Need

- A real-time update dashboard for appointments, bed/room availability and staff schedules
- Automated report generation for financial, staff workload and patient statistics.
- User management tools (create, edit, delete accounts, set access levels).
- Security features to protect the system from unauthorized access
- Alerts for emergencies.
- Notification for schedule clashes between the hospital staff.
- Link to the billing and insurance modules.

Influence

- High – sets requirements and validates system performance.
- Can approve or reject design decisions (features, workflows).
- Influences training and adoption among hospital staff.
- Acts as the main decision-maker during system rollout.

Relationship

- Direct and daily use – Admin will log in multiple times a day.
- Relies on the system to plan, monitor, and control hospital resources.
- Controls the efforts put into the document as well as the efforts by the hospital personnel.
- Takes charge of the organization after the system introduction.

PERSONA

Name: Owen Leong Role: Hospital Admin	
Characteristics	35 years old. Owen has been managing hospital operations for over 15 years. His responsibilities cover staffing, patient admissions, finance, compliance, and ensuring smooth coordination between departments.
Core Goals	Maintain smooth day-to-day operations, ensure patient safety and staff productivity, reduce administrative errors (e.g., double-booking, billing mistakes), access real-time reports on hospital capacity and performance and ensure compliance with government regulations and data security standards.
Typical Challenges	Faces delay due to fragmented or manual processes and struggles to monitor multiple departments at once.
Singularities	Owen has the authority to oversee all hospital operations. Besides that, he balances both patient care quality and administrative efficiency.
Main Task with System Support	Schedule doctors, allocate patient rooms, monitor hospital resources (beds, equipment, staff availability), ensure billing accuracy, generate compliance/performance reports and sent alerts for emergencies
Miscellaneous	Owen works in high-pressure peak periods in the hospitals works for very long periods, and depends a lot on rapid and highly precise data for decision-making. He wishes for a reliable and straightforward system that minimizes manual

	work, provides uninterrupted communications between personnel, physicians, and patients, and avoids patient mistrust and exposure of sensitive information.
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Medical Staff

Goals

- Provide high-quality and empathetic care to patients.
- Record correct daily observations and personal notes for patients.
- Access patient profiles and histories to support medication decisions.
- Respond to patient communications promptly.
- Track patient well-being trends for better monitoring.

Role

- They document clinical observations during daily rounds
- They review and respond to patient messages.
- They use the system to access appointments and prepare for patient interactions

Need

- Secure staff accounts with unique IDs and role-based access.
- Tools to quickly log daily notes and update patient information.
- Ability to view cultural, language, and personal preferences of patients.

Influence

- Medical staff are important because they are responsible for maintaining the accuracy and completeness of patient data.
- Their influence is also high, as their feedback helps refine workflow efficiency, usability, and clinical accuracy of the system.

Relationship

- Medical staff interact with the system daily to record and review information.
- The system acts as a digital patient diary, communication hub, and monitoring tool for staff.
- Staff rely on the system for accurate, up-to-date records to make care decisions and ensure patients' preferences are respected

Name: Emily Role: Hospital Medical Staff	
Characteristics	Emily has over ten years of experience in patient care and is known for her empathetic and patient-centered approach. While she is comfortable with technology, she is not deeply technical, so she values systems that are simple, intuitive, and quick to use.
Core Goals	Provide high-quality, empathetic care by accurately recording daily observations and personal notes, accessing patient information efficiently, and ensuring that patients' medical, emotional, and cultural needs are respected.
Typical Challenges	Her main challenge is the time pressure she faces during busy ward rounds. Paper-based notes, which must later be transcribed into official records, are a constant frustration for her.
Singularities	Emily is unique because she always remembers small details about her patients, such as their food preferences, cultural sensitivities, or morning routines.
Main Task with System Support	She relies on the system to quickly access patient profiles and scheduled appointments so she can prepare for her rounds. She also uses the system to search for patients by name, ID, or contact number, which helps her manage multiple patients efficiently.
Miscellaneous	Emily values systems that help reduce paperwork and duplication, as she dislikes repeating tasks unnecessarily

Doctor

Goals

- Deliver accurate diagnoses and effective treatment plans for patients.
- Ensure continuity of care across multiple visits and referrals.
- Review patient history, lab results, and diagnostic images efficiently.
- Communicate clearly with patients and other hospital staff.
- Track patient treatment outcomes to improve medical decision-making.

Role

- They interpret medical test results and document diagnoses.
- They prescribe medications and treatment regimens for patients.
- They coordinate with other staff and specialists for integrated care.
- They use the system to prepare for consultations and follow up on treatment plans.

Need

- Secure access to detailed patient medical history and test results.
- Decision-support tools (e.g., drug interaction alerts, clinical guidelines).
- Easy access to referral and appointment scheduling.
- Quick review dashboards to track chronic conditions and progress.

Influence

- Doctors have very high influence as they make core clinical decisions.
- Their feedback significantly impacts system design for usability, speed, and decision support accuracy.
- They shape workflows that other staff and patients will follow.

Relationship

- Doctors use the system daily for patient consultations, record reviews, and prescriptions.
- The system acts as a central medical knowledge and record hub.
- Doctors rely on the system for accurate, consolidated, and up-to-date information to ensure safe and effective treatment.

Name: Dr James Wong	
Characteristics	48-year-old physician. He is a person who is highly knowledgeable, empathetic and values efficiency. He is professional and confident with medical technology but often under pressure due to the workload and patient volume.
Core goals	Deliver accurate diagnoses and effective treatments in a timely manner. He aims to balance clinical excellence with patient-centered care, ensuring the patients understand their conditions and treatment options.
Typical challenges	Struggles with limited time per patient, the overwhelming amount of clinical documentation and occasional resistance from patients who distrust medical advice. Language barriers and cultural differences can also make communication difficult. In addition, he sometimes cannot access a patient's full medical history due to fragmented records and privacy restrictions.
Singularities	He prefers structured, evidence-based data at his fingertips. He values decision-support tools that reduce diagnostic uncertainty. He also strongly believes in multidisciplinary collaboration with nurses and specialists.
Main task with System Support	Use the CareLog system to review patient histories, monitor lab results, prescribe medications and enter clinical notes. He also receives alerts for critical patient conditions that need immediate attention.
Miscellaneous	He values systems that reduce administrative burden. Outside of work, Dr Wong is passionate about medical education and occasionally contributes to research.

Nurse

Goals

- Provide bedside care and monitor patient recovery.
- Support doctors by executing treatment plans and administering medications.
- Maintain detailed and up-to-date patient charts.
- Ensure patients' comfort, safety, and well-being.
- Act as a bridge between doctors, patients, and families.

Role

- They record vital signs, medication administration, and shift notes.
- They prepare patients for procedures and assist during medical interventions.
- They use the system to update patient care records and check treatment instructions.
- They communicate patient concerns to doctors and other staff.

Need

- Quick and simple note-taking tools for shift reports.
- Medication tracking and alerts for dosage/timing.
- Access to care protocols and patient-specific instructions.
- Secure, role-based login for data protection.

Influence

- Nurses strongly influence system usability, as they interact frequently and in time-sensitive contexts.
- Their feedback ensures the system supports fast, accurate documentation and minimizes errors.
- They provide critical insights into patient day-to-day care.

Relationship

- Nurses use the system continuously during shifts for updates and documentation.
- The system acts as a real-time care tracker and coordination hub.
- Nurses rely on the system to ensure proper execution of doctors' orders and to record accurate patient progress.

Role: Nurses

Name: Sarah Lim	
Characteristics	34-year-old nurse. She is compassionate, detail-oriented and highly approachable to patient. She adapts quickly in fast-paced environments and is skilled at handling both medical and emotional needs of patients.
Core goals	Ensure patients feel safe and cared for, while supporting doctors in implementing treatment plans. She aims to maintain accurate records, provide timely medication administration and advocate for patients' comfort and dignity.
Typical challenges	Difficult to maintain clear communication between doctors, fellow nurses and patients, especially during shift changes or emergencies. She sometimes struggles to quickly understand whether a patient's condition is improving or deteriorating, since updates are not always shared promptly or recorded consistently.
Singularities	She builds strong rapport with patients by listening attentively. Sarah often becomes the first point of contact for patient concerns and helps bridge communication gaps between the doctors and patients.
Main task and support	Use the Carelog system to log vital signs, track medication schedules, update daily care notes and check shift handover details. She also uses it to set reminders for time-sensitive procedures.
Miscellaneous	She values teamwork and appreciates when digital tools streamline coordination between departments. Sarah often volunteers for mentorship of new nurses and is motivated by patient appreciation

Psychological Counsellor

Goals

- Support patients' mental health through therapy and counselling.
- Maintain confidential, accurate notes on sessions.
- Track patient emotional and behavioural progress over time.
- Provide personalized support based on cultural and personal preferences.
- Encourage patient engagement in treatment plans.

Role

- They conduct therapy sessions and document observations.
- They respond to patient communications regarding mental well-being.
- They use the system to track therapy schedules, session notes, and patient progress.
- They collaborate with doctors and nurses when psychological issues affect medical care.

Need

- Highly secure, private record-keeping to protect patient confidentiality.
- Tools to log session summaries, emotional status, and therapeutic progress.
- Access to patient preferences, backgrounds and care history.
- Communication features to check in with patients between sessions.

Influence

- Counsellors' influence is moderate to high, as their feedback helps refine sensitive communication and privacy workflows.
- They provide insights into mental health tracking tools and patient-centered design.
- Their input ensures the system respects cultural, emotional, and ethical considerations.

Relationship

- Counsellors interact with the system regularly but less frequently than nurses or doctors.
- The system serves as a private therapy log, communication tool, and monitoring system.
- Counsellors rely on the system for accurate, secure documentation and holistic views of patient well-being.

Role: Psychological Counsellor

Name: Aisha Wong	
Characteristics	39-year-old. Aisya is empathetic, patient and skilled in supporting patients' mental health. She is comfortable with technology but prefers personal conversations over digital tools. She collaborates closely to nurses and doctors to integrate mental health into overall treatment.
Typical challenges	Faces patients who are reluctant to open up, struggles with incomplete emotional health records and sometimes experiences scheduling conflicts with medical staff. She also finds it difficult when systems don't integrate psychological notes with patient medical data.
Singularities	She values private, secure communication channels, prefers systems that allow confidential notes and depends on nurses to flag patients who may need mental health support.
Main task and support	Use the hospital system to review patient history, log counselling sessions and coordinate with nurses and doctors about patients' emotional states and treatment adjustments.
Workplace	Her "workplace" is in private counselling rooms and patient wards when bedside support is required.
Main Important Task	Providing timely psychological support and ensuring patients' mental health needs are integrated with their overall treatment plans.

Patient

Goals

- Patients can be involved in their own care
- Record daily health information
- Ensure personal preferences are respected by staff
- Privacy being protected and controlling themselves via system
- Communicate with the doctor, medical staff and nurse easily and efficiently

Role

- Patients are the main users of the system
- Patients input data every day
- Patients used the features such as booking an appointment, creating a reminder and messaging staff

Need

- Patients have an account with a user ID and password
- Patients can manage their privacy settings
- Patients can receive notifications for medication, appointments

Influence

- Patients are highly important as they are the main reason for this system to exist
- Patient influence is also high as patients' feedback shapes the system feature, design and UI.

Relationship

- Patients interact directly and regularly with the system. Example, the reminders, daily survey and appointment)
- The system plays the role of a health diary for the patient, a communication tool between the patient and staff and patient preferences.

Name: Nicole Tan Role: Hospital Long-Term Care Patient	
Characteristics	65-year-old. Nucle Tan is a kind person. She occasionally finds digital systems confusing because she is not very tech-savvy. Being a non-native English speaker, she frequently favors speaking in her native language
Core Goals	Feel involved in her healthcare to follow the medication schedule correctly and ensure that her personal preferences and cultural beliefs are respected.
Typical Challenges	Has trouble remembering when to take her medications, struggles with language barriers and feels overloaded with medical information. Additionally, she finds it stressful when her privacy is not completely protected
Singularities	She prefers female nurses. She wants meals to be in line with her dietary preferences
Main Task with System Support	Nicole Tan records her daily pain level, mood and sleep quality using the CareLog system. She also checks reminders for taking medicine on time.
Miscellaneous	Nicole Tan places a high value on privacy and frequently decides to keep some medical information private from non-staff who are not necessary. She feels safer and comfortable when she can control access to information.

Interview questions and answers (Patient)

What information would you prefer to record every day?	I like to record how much pain I feel, how well I slept, and my mood. Sometimes I also write a little about how I'm feeling emotionally, such as when I feel lonely or worried.
How would you like to be notified if you have an appointment at the hospital?	I prefer receiving a message on my phone — something simple, perhaps with a picture or a voice message. If it's only in English, sometimes I get confused. If it could be in my language, that would be much better.
Feelings when sharing health information with the staff?	If I know the staff well and trust them, I feel okay. But sometimes I worry about who can see my information. I don't want everyone to know everything, especially things that are personal.
Your personal preferences?	I prefer female nurses — I feel more comfortable with them. I also like my meals to follow my dietary needs, like less oil and no pork. I enjoy quiet time in the afternoon, and I don't like being rushed when I talk.
Prefer giving feedback through the system?	Sometimes, yes — if it's easy to use. But if the system is too complicated, I get frustrated. I prefer talking to someone face-to-face when I have something important to say. Maybe the system can help me write short feedback, and then someone can follow up with me.
Keep track of your sleeping quality, mood or pain levels to determine if they are good for you. Why?	Yes, it helps me understand my body better. When I notice that my pain is worsening or I'm not sleeping well, I can tell the nurse or doctor. It also makes me feel more in control.
Have religious beliefs you hope the staff remembers?	Yes, I don't eat certain foods because of my beliefs. I also prefer quiet time for prayer in the evening. I feel respected when staff remember these things.
Do you prefer to send messages or speak face-to-face?	I prefer face-to-face. I feel more connected when I can see someone's face and hear their voice. But if they are busy, I don't mind sending a message — as long as it's simple and someone replies soon.

4. System Features and Requirements

Patient	
System Features	
In scope	- Patient could create and manage their own account, profile with emergency contact and medical details - Patient can record mood, pain level, sleep quality, and medication status with history - Patients have the channel to send messages to staff and receive feedback from the staff - Patient can book the appointment - Patient can update their food/ language/ nurse/gender preferences
Out-of-scope	- Patient set visibility levels for personal details and medical history
Functional Requirements	
In scope	- The system shall allow patients to register or log in via their phone number or email with a password and the verification safety question - The system shall allow password reset by the patient via a verification safety question - The system shall allow patients to set their food/ nurse/ gender/ language preferences - The system shall allow patients to update their gender, medical details, and emergency contact information - The system shall provide a daily survey - The system shall store login history - The system shall allow patients to send messages to staff - The system shall allow patients to submit feedback - The system shall allow patients to book appointments
Out-of-scope	- The system shall allow patients to schedule reminders - The system shall display bookings in calendar view - The system shall allow patients to hide their medical details
Non-Functional Requirements	
In scope	- The system shall support multi-factor authentication for sensitive operations (Security) - The system shall enforce role-based access so patients can hide data from non-relevant staff (Availability)
Out-of-scope	- The system shall log all access attempts for compliance monitoring. (Security) - The system shall ensure no data is lost via daily backups. (Reliability) - The user interface shall support at least 3 languages. (Usability)

	- The system shall provide accessibility features such as a large text option or contrast mode. (Usability) - The system shall run on major platforms. (Portability)
Acceptance Criteria	
In scope	- The system shall allow patients to hide medical details from non-primary staff (Privacy) - The system shall allow patients to edit their personal information in the profile (Privacy) - The system shall allow patients to submit feedback (Communication) - The system shall allow patients to set their own reminders in the calendar (Medication) - The system shall allow patients to view the reminder in the calendar (Medication) - The system shall allow patients to create their own account with a unique user ID and password (Patient profile) - The system shall allow patients to edit their personal information and make changes (Patient profile) - The system shall lock the accounts after 3 failed login attempts (Patient profile)
Out-of-scope	- The system shall allow patients to switch language options - The system shall provide patients with the data visibility summary - The system shall send patients the medication reminder before the scheduled time

Medical Staff	
System Features	
In scope	- Medical staff can log into the system with secure authentication. - Medical staff can record daily observations about patients. - Medical staff can receive and respond to patient messages - Medical staff can view appointments booked by patients. - Medical staff can get the doctor's diagnostic record before giving the medical - Medical staff can view the patient's details and medical record to prevent the patient's allergies when giving medical.
Out-of-scope	- Medical staff can view patient profiles, including medical history, emergency contacts, and preferences. - Medical staff can have suggestions when giving medical care to the patient

	- Medical staff can have an alert or reminder about the effect of the medication due to the condition of the patient - Medical staff can follow up on the patient's details to evaluate how this medication affects the patient
Functional Requirements	
In scope	- The system shall allow medical staff to log in using their staff ID and password with secure authentication. - The system shall allow medical staff to record daily observations about patients (mood, appetite, medication, pain, sleep quality). - The system shall allow medical staff to view a list of all patients assigned to them. - The system shall allow medical staff to view appointments scheduled by patients. - The system shall enable books, rescheduling and cancelling patient appointments. - The system shall provide search functionality to find patients using ID, name or contact number.
Out-of-scope	- The system shall allow medical staff to receive and reply to messages from patients. - The system shall allow medical staff to create or delete patient accounts.
Non-Functional Requirements	
In scope	- The system shall require medical staff to use multi-factor authentication for login. (Security) - The system shall enforce role-based. (Security) - The system shall provide a user-friendly interface. (Usability)
Out-of-scope	- The system shall support offline functionality for staff. (Availability) - The system shall support voice-to-text dictation for staff notes. (Usability)
Acceptance Criteria	
In scope	- The system shall allow medical staff to log in using their staff ID and password. (Authentication) - The system shall lock medical staff accounts after 3 failed login attempts. (Authentication) - The system shall automatically timestamp all staff entries. (Patient Log Management) - The system shall allow medical staff to receive messages from patients. (Communication)
Out-of-scope	- The system shall allow medical staff to reply to patient messages. (Communication)

	- The system shall allow medical staff to create daily observation logs for assigned patients. (Patient Log Management)
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Doctor	
System Features:	
In scope	- Create and manage the doctor's profile (specialty, qualifications, license no., working hours, contact) - View assigned/consented patient summaries, allergies, meds, results and prior encounters - Create or update the encounter notes and diagnosis; add orders/recommendations (labs, imaging, medication instructions) with version history. - Upload and view clinical attachments (labs, scans, letters) linked to the encounters. - Manage appointments (see agenda, accept/reschedule/cancel); set clinic availability and breaks
Out-of-scope	- Telehealth video /screen-sharing - eRx to external pharmacies and controlled-substance workflows. - Billing /claims/insurance and price estimates - Organization-wide role/ permission administration and audit policy configuration - Two-way secure messaging with patients and staff , cirt-events alerts - Bulk chart export/BI dashboards; data warehousing - External calendar sync. (Google/Outlook) and on-call rota management
Functional Requirements	
In scope	- The system shall allow doctors to register /login via phone/email with a password and verification safety question - The system shall allow password reset via verification of safety questions - The system shall allow editing of professional profiles (specialty, qualifications, hours, contact). - The system shall display patient conditions, allergies, medications, results and history within permissions - The system shall allow creation/ update of notes and diagnosis with timestamps and author ID; prior versions are retained - The system shall display patient conditions, allergies, medications, results and history within permissions - The system shall allow entering orders/ recommendations as chart entries tied to encounters - The system shall provide secure messaging with patients/staff and allow marking threads as resolved

	- The system shall allow accepting/rescheduling/cancelling appointments and viewing a real-time schedule with conflict prevention - The system shall store login history for each doctor account
Out-of-scope	- Doctors altering patient demographic master data (name/DOB/identifier) - Doctors configuring system-wide templates, roles or notification rules - Printing/signing legal certificates (e.g; sick leave, death certificate) within the app) - Custom analytics builder (queries, cohorts, dashboards)
Non-Functional Requirements	
In scope	- Security: RBAC, consent gate on sensitive modules, data encryption in transit /at rest, auto-logout after inactivity, optional MFA for sensitive actions - Auditability: Log access and clinical edits with timestamp, user ID, patient ID, reason (where applicable) - Performance: Load patient record and schedule within ~ 2 seconds under normal load - Availability: ≥ 99.9 % uptime during operating hours - Interoperability: Integration points for labs/pharmacy/imaging to surface results; failures degrade gracefully - Portability/usability: Responsive web app; tablet-friendly for bedside boarding 2 hours - Maintainability: Structured logging/metrics; health checks
Out-of-scope	- Formal regulatory certification statements - Disaster recovery targets beyond standard backups - Multilingual UI and advanced accessibility (screen readers) in this phase
Acceptance Criteria	
In scope	- Access control: opening a non-assigned patient's history record is blocked with a consent error; event is logged - Profile: doctor can create an account ;verify identity and edit professional details; admin-only fields remain locked - Security: account locks 3 failed logins and requires reset - Scheduling: double booking is prevented; appointments appear in calendar view; booking/ cancelation notifies doctor and patient; cancelled slots re-open automatically.

	- Clinical documentation: doctor can add/edit notes and diagnoses; each change shows timestamp/author; a prior version can be viewed - Messaging: doctor replies to patient messages; thread can be marked resolved and archived
Out-of-scope	- Telehealth: starting video visits from the schedule - eRx: sending prescriptions to pharmacies - analytics: cross patient dashboards



Nurse	
System Features:	
In scope	- Execute treatment plans, collaborating with the doctor, psychological counsellor and medical staff - Record vital signs; log in administer medications (time/amount/route); track symptoms/pain/progress - Update daily care notes and nursing assessments; review historical nursing logs - Receives notification when doctor updates orders or a patient requests help - Share the latest patient status and flag urgent conditions
Out-of-scope	- Entering diagnoses or prescribing medications independently - Charging doctor schedules or clinic availability - Editing patient demographic master data - Integrating bedside - Managing billing payment or insurance tasks
Functional Requirements	
In scope	- The system shall allow nurses to log medication administrations (drug, dose ,route, time) with nurse ID. - The system shall allow nurses to record vital signs and observations; attach assessments to encounters - The system shall provide a prioritized daily task list; tasks can be marked completely. - The system shall provide real-time access to current treatments plans and historical nursing documentation - The system shall notify nurses on updated doctor orders or patient calls; nurses can request clarifications - The system shall allow secure messages with doctor and staff; urgent cases can be flagged
Out-of-scope	- Nurses modifying doctor orders without a counter-signature workflow - Creating/editing organization-wide templates/form/notification rules - Bulk exporting nursing records
Non-Functional Requirements	
In scope	- Security & privacy: RBAC; encryption in transit/at rest; auto-logout; optional MFA for medication administration

	- Performance: Retrieve /update nursing record within~ 2 seconds - Audibility: Log all nursing activities with timestamp and nurse ID - Usability/Portability: Tablet-first UI; ≤2 hours training for new nurses.
Out-of-scope	- Reliability/availability: ≥99.9% uptime during operation time for record access. - Interoperability: Surface lab/ pharmacy/ imaging results referenced by doctor orders Formal compliance guarantees/certifications text - Performance: Retrieve /update nursing record within~ 2 seconds - Scalability: Support ≥200 concurrent nursing users with target latencies. - Storability: Disaster recovery objectives beyond standard backups
Acceptance Criteria	
In scope	- Medication admin: A nurse records an administration and sees it on the MAR; entry is timestamped with nurse ID. - Vital & notes: Vital/notes save to the timeline and are reviewable; edits display timestamp and author - Tasks: The daily task list show due times ; tasks change to “Completed” when recorded ; high-risk patient are highlighted - Notifications: Doctor order updates or patient calls trigger a notification within seconds that can be acknowledged - Access control: Attempts to open a non-assigned patient record are blocked and logged - Security: Account locks after 3 failed login attempts
Out-of-scope	- Prescribing: creating medication orders - Scheduling: managing doctor availability or clinical sessions - Billing: viewing or altering billing and payments

Psychological counsellor	
System Features:	
In scope	- Create & manage counsellor profile (license, specialties, clinical interests, working hours, contact) - View consented clinical summaries and risk flags; limited read of meds/allergies for safety; no unrestricted access to full medical history - Record counselling session notes, therapy plans, goals, and outcomes; track standardized scales with history - Two-way secure messaging with patients (non-emergency); set expectations and boundaries in-app - Manage counselling appointments (individual and ,optionally,group sessions) and set availability - Escalation channel to on-call doctor/nurse for crisis situations; document safety plans and next steps
Out-of-scope	- Medical diagnosis and prescribing - Editing doctor orders or treatment instructions - Accessing highly sensitive categories (e.g., reproductive health, HIV results) without explicit consent or break-glass justification - Telehealth video; external resource referrals with automated document sharing - Issuing legal attestations/ letters to external entities from within the app -
Functional Requirements	
In scope	- The system shall allow counsellors to register/login, reset their password via safety question, and edit professional profile - The system shall restrict access to patients based on assignment and consent; sensitive modules require explicit consent or break-glass with reason - The system shall provide counsellor-specific session note templates (SOAP/DARE) and allow attaching worksheets/resources - The system shall capture standardized assessments and auto-compute scores and severity bands; scores appear on the patient timeline - The system shall allow creating/updating therapy goals and safety plans; follow-up tasks/reminders are created for patient (viewable to patient and counsellors) - The system shall provide secure messaging with patients; out-of-hours or crisis keywords trigger an auto-response with emergency guidance and escalate to staff - The system shall allow booking/scheduling/canceling counselling sessions; calendar reflects changes in real time and prevents double booking

Out-of-scope	- Bulk export of therapy notes to third parties - Organization-wide template/notification configuration by counsellors
Non-Functional Requirements	
In scope	- Privacy by design: Therapy notes are stored in a protected area; access requires assignment and consent; redaction of sensitive fields in general views. - Security: RBAC, encryption, auto-logout; MFA for break-glass access
Out-of-scope	- Auditability: All access to therapy notes is logged with reason when break-glass is used - Performance: Assessments compute and render within ~1 second; session notes save within ~ 2 second - Availability: ≥99.8% during operating hours. - Usability: Clear boundaries messaging (“not for emergencies”);mobile friendly for private use - Multilingual UI and advanced accessibility in this phase
Acceptance Criteria	
In scope	- Session notes: Counsellor creates a session note using the template; saving retains timestamp, author and version history - Safety plans: counsellor can document a safety plan with triggers, coping strategies , and contacts; patient can view their copy - Messaging boundaries: messages containing crisis keywords trigger an immediate auto reply with emergency guidance and notify on-call staff - Scheduling: double booking is prevented: sessions appear in calendar view; booking/cancellation notifies counsellor and patient - Security: account locks after 3 fail login attempts
Out-of-scope	- External attention: generating official letters for schools/employers from within the app

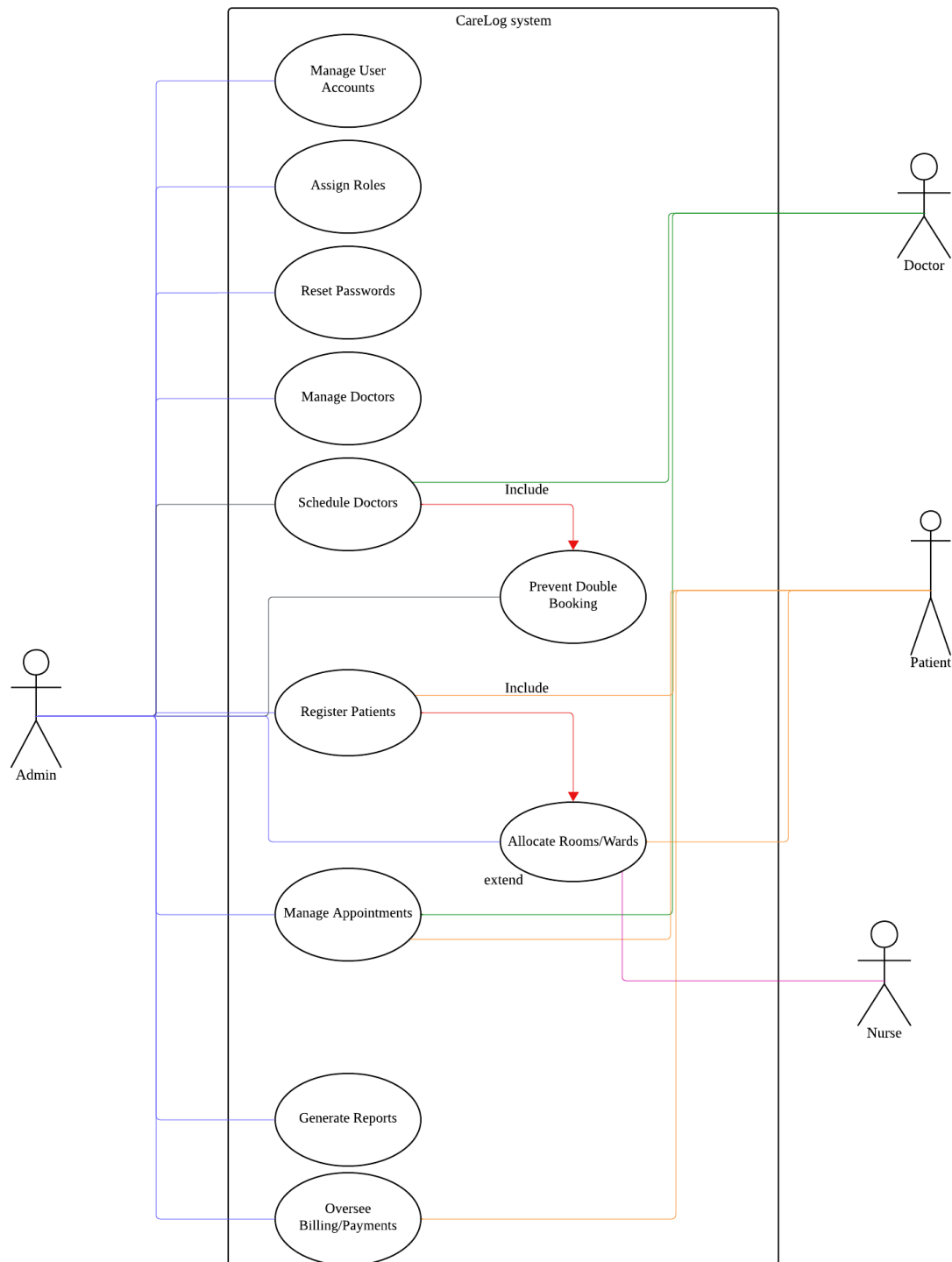
Hospital Admin	
System Features	
In scope	User Management: – The system manages staff and patient accounts with secure role-based access and password recovery. Doctor Management: – The system enables administrators to add, update, schedule, and manage doctors without conflicts. Patient Management: – The system registers patients, maintains medical history, and ensures secure record access. Appointment & Scheduling: – The system allows booking, updating, and canceling appointments while preventing overlaps. Room & Resource Allocation: – The system tracks real-time room and equipment availability to ensure efficient allocation. Billing & Payment: – The system generates bills, processes payments, and integrates with insurance securely. Reporting and Analytics: – The system produces real-time reports and dashboards for hospital performance monitoring. Communication Module: – The system provides secure messaging, reminders, and emergency notifications for staff and patients.
Out-of-scope	Billing & Payment Integration: – Integration with external insurance APIs for claim processing is out of scope as it requires third-party collaboration and advanced setup beyond this project. Communication Module (Secure Messaging): – End-to-end encrypted real-time messaging between staff is out of scope since it involves complex networking and security frameworks not covered in this project.
Functional Requirements	

In scope	- The system shall register hospital staff (doctors, nurses) - The system shall update hospital staff (doctors, nurses) - The system shall remove hospital staff (doctors, nurses) - The system shall enforce role-based access control - The system shall allow password reset and account recovery - The system shall maintain and update doctors' profiles (specialization, availability, qualifications) - The system shall schedule and manage doctors' timetables for consultations and surgeries. - The system shall prevent conflicts such as double booking. - The system shall register new patients with personal and medical details - The system shall track patient history, treatment records and appointments. - The system shall allocate patients to appropriate wards or rooms. - The system shall book, update or cancel patient appointments. - The system shall notify patients and doctors of scheduled visits. - The system shall manage operating theatre bookings and ensure no overlap - The system shall monitor availability of hospital rooms, wards and operation theatre - The system shall allocate rooms and equipment efficiently based on patient needs. - The system shall generate bills for patient services and treatments - The system shall record payments and outstanding balances. - The system shall generate reports on patient admissions, treatments, billing and resource usage. - The system shall provide dashboards for administrators to monitor hospital performance. - The system shall send alerts, reminders or emergency notifications to specific staff. - The system shall allow secure communication between doctors, nurses and administrators.
Out-of-scope	- The system shall allow email/SMS-based recovery or multi-factor authentication. - The system shall integrate with insurance claim systems if needed.
Non-Functional Requirements	
In scope	- The system shall respond to user requests within 2 seconds - The system shall implement data encryption for sensitive information - The system shall use strong authentication and role-based authorization - The system shall have a user-friendly interface that requires minimal training--

	- The system shall have automatic data backup - The system shall support future expansion - The system shall provide logging and monitoring for troubleshooting
Out-of-scope	- The system shall handle at least X concurrent users without crashing - The system shall ensure compliance with healthcare data privacy regulation - The system shall have accessibility features for the visually impaired user - The system shall support multi-language options for diverse staff and patients - The system make sure 99.9% system uptime - The system shall have disaster recovery. - The system shall accessible via web browsers and mobile
Acceptance Criteria	
In scope	User Management: - The system shall allow administrators to create, update and delete staff accounts. Doctor Management: - The system shall allow administrators to add new doctors with details (name, specialization, qualifications, availability). Patient Management - Administrators shall be able to register patients with personal and medical details. Appointment & Scheduling - The system shall allow patients and admins to create, update, and cancel appointments. Room & Resource Allocation: - The system shall display real-time room availability. Billing & Payment: - The system shall automatically generate bills based on patient services and treatments. Reporting and Analytics: - The system shall generate reports (billing, patient admissions, doctor schedules) within 10 seconds. Communication Module: - The system shall allow secure messaging between administrators, doctors, and nurses.
Out-of-scope	Patient Management: -Integration with external Electronic Health Records (EHR) systems. Billing & Payment: - Integration with external payment gateways (e.g., PayPal, Stripe).

5. Use Case Diagram

Admin



(if the pic is too blurry, please click into the link for this chart in the references)

Description

Primary Actor:

-**Admin** (main controller of staff, patients, billing, resources).

Secondary Actors (interact with Admin features):

-**Doctor** → receives schedules, appointments, and notifications from Admin.

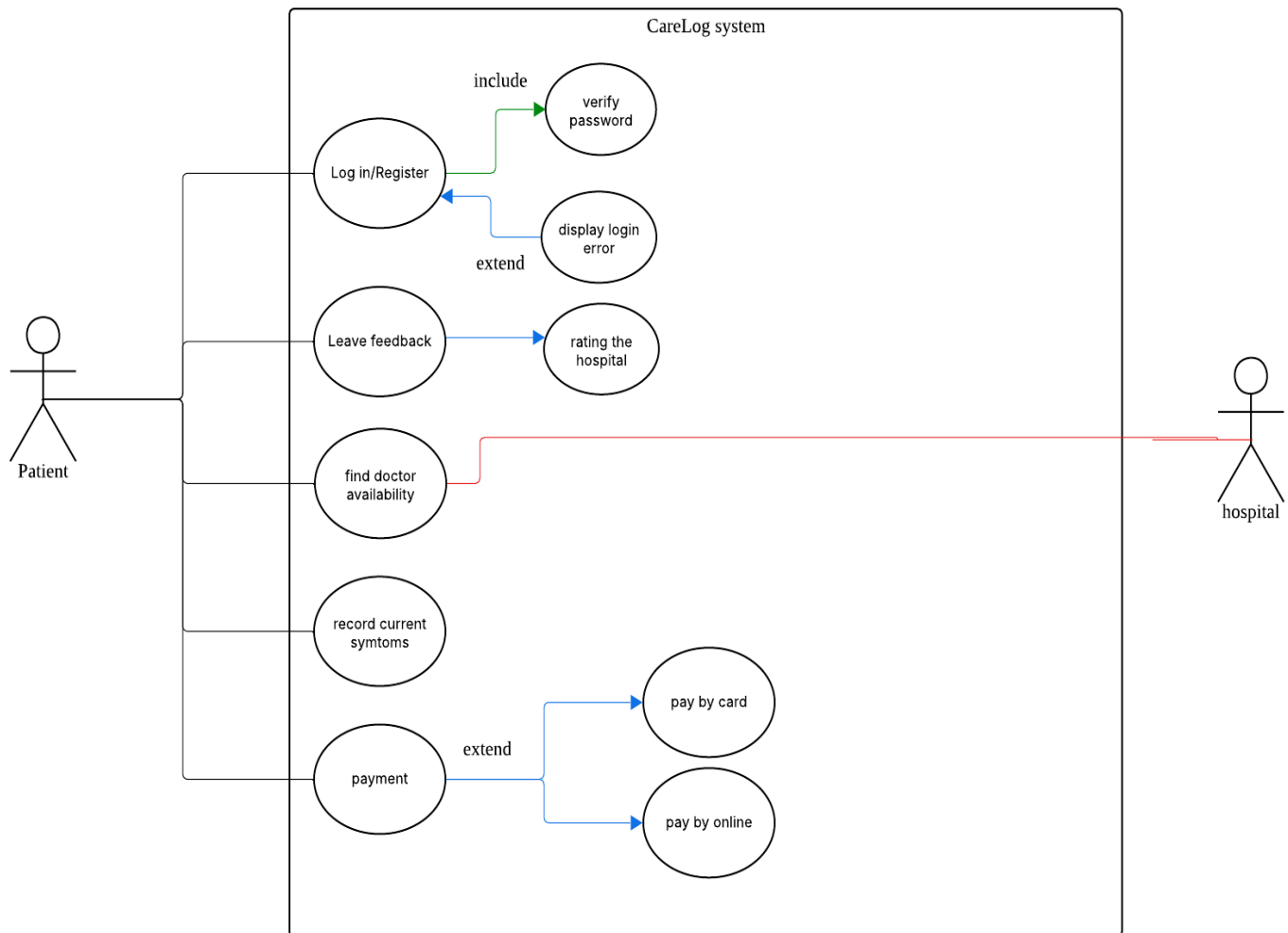
-**Nurse** → receives room allocations, shift assignments, and emergency alerts.

-**Patient** → registered by Admin, allocated rooms, billed, and scheduled for appointments.

Admin Use Case Diagram

The Admin Use Case Diagram highlights the role of the hospital administrator In interfacing with the system to handle vital hospital functions. The Admin is the central actor and is in charge of managing users and staff, admitting and scheduling patients, assigning rooms and resources, preparing bills, and handling various reports. As for secondary actors, there are doctors, nurses, patients, and the billing staff who are to varying degrees of peripheral relevance to the Admin's work due to case file scheduling, transaction notifications, resource and staff allocation, and billing activities. The diagram covers functions like resolution of resource conflicts, report generation, and notifications to the users, which are essential from the point of balance in workflow efficiency, security, and coordination in the hospital.

Patient Use Case Diagram



Primary Actor:

- **Patient** (central actor who receives care, schedules visits, and communicates needs).

Secondary Actors (interact with patient features):

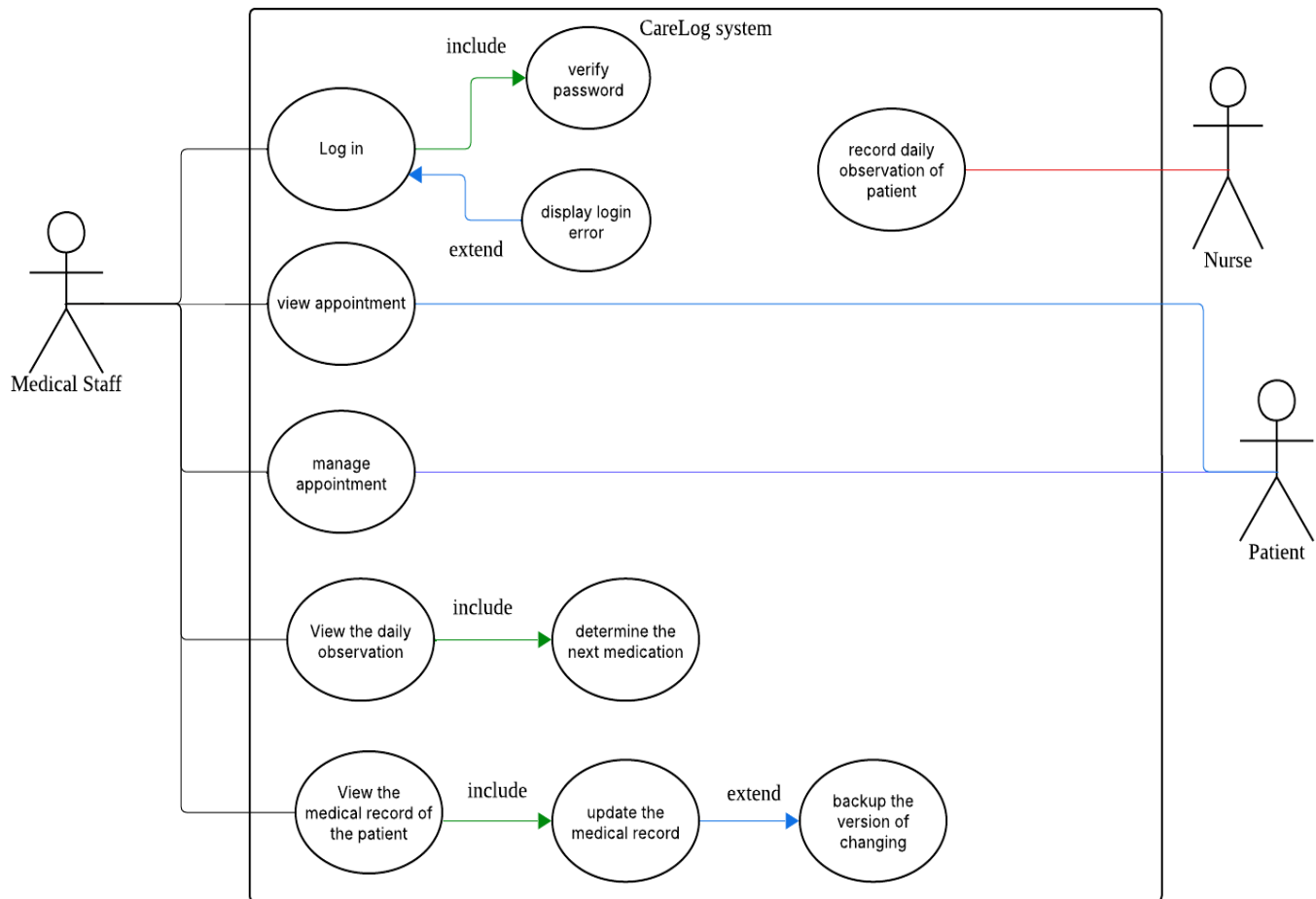
- **Doctor** → provides diagnoses, treatment, and follow-ups.
- **Nurse** → records vitals, administers medications, and updates care logs.
- **Medical Staff** → documents observations, manages daily notes, and monitors well-being trends.
- **Psychological Counsellor** → supports patient mental health through counselling sessions and therapy notes.
- **Admin** → registers patients, allocates rooms, manages appointments, and handles billing.

Description:

The Patient Use Case Diagram highlights the role of patients in actively engaging with the system for their healthcare journey. Patients are the central actors who interact with doctors, nurses, counsellors,

and medical staff for appointments, treatment, and mental health support. Patients receive schedules, personalized care, and therapy while also being billed and registered by the admin. Their role emphasizes communication, sharing preferences, and accessing their medical history, which supports continuity of care and collaborative decision-making.

Medical Staff Use Case Diagram



Primary Actor:

- **Medical Staff** (main actor responsible for documenting daily observations and maintaining patient records).

Secondary Actors:

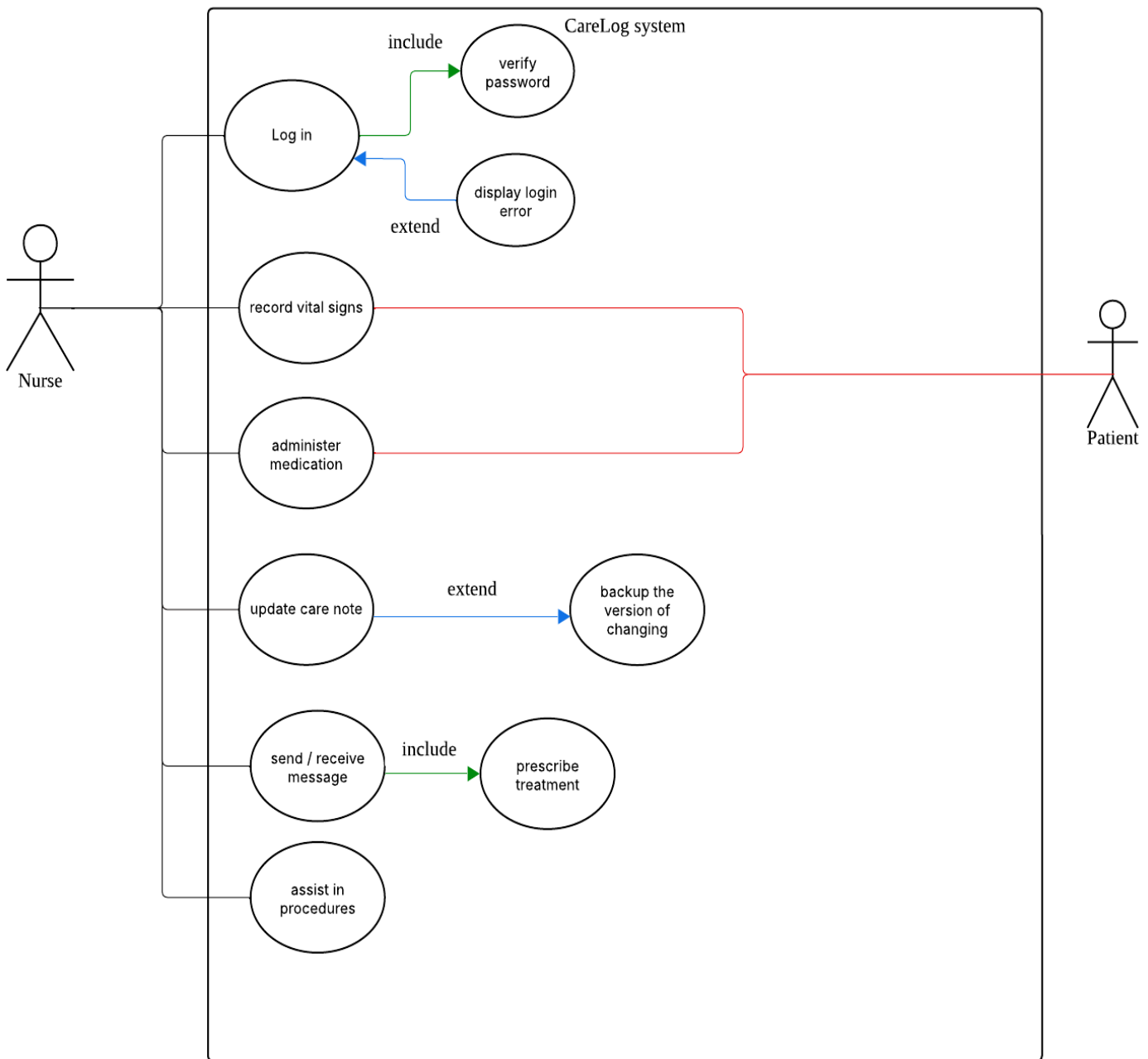
- **Doctor** → relies on staff notes for diagnosis and treatment planning.
- **Nurse** → coordinates with staff to ensure accuracy of daily logs and patient progress.
- **Patient** → receives empathetic care and benefits from accurate records.
- **Psychological Counsellor** → accesses relevant history and well-being trends to provide therapy.
- **Admin** → grants access permissions and manages accounts for staff.

Description:

The Medical Staff Use Case Diagram highlights the role of medical staff in maintaining accurate and timely patient records. They act as daily observers, recording vital notes, symptoms, and patient feedback into the system. They also manage communications with patients and share updates with nurses and doctors to support clinical decisions. Their role is central to ensuring that the system reflects

real-time patient status, helping doctors and counsellors make informed choices and ensuring patients' cultural and personal preferences are respected.

Nurse Use Case Diagram



Primary Actor:

- Nurse (key actor providing bedside care and monitoring patient progress)

Secondary Actors:

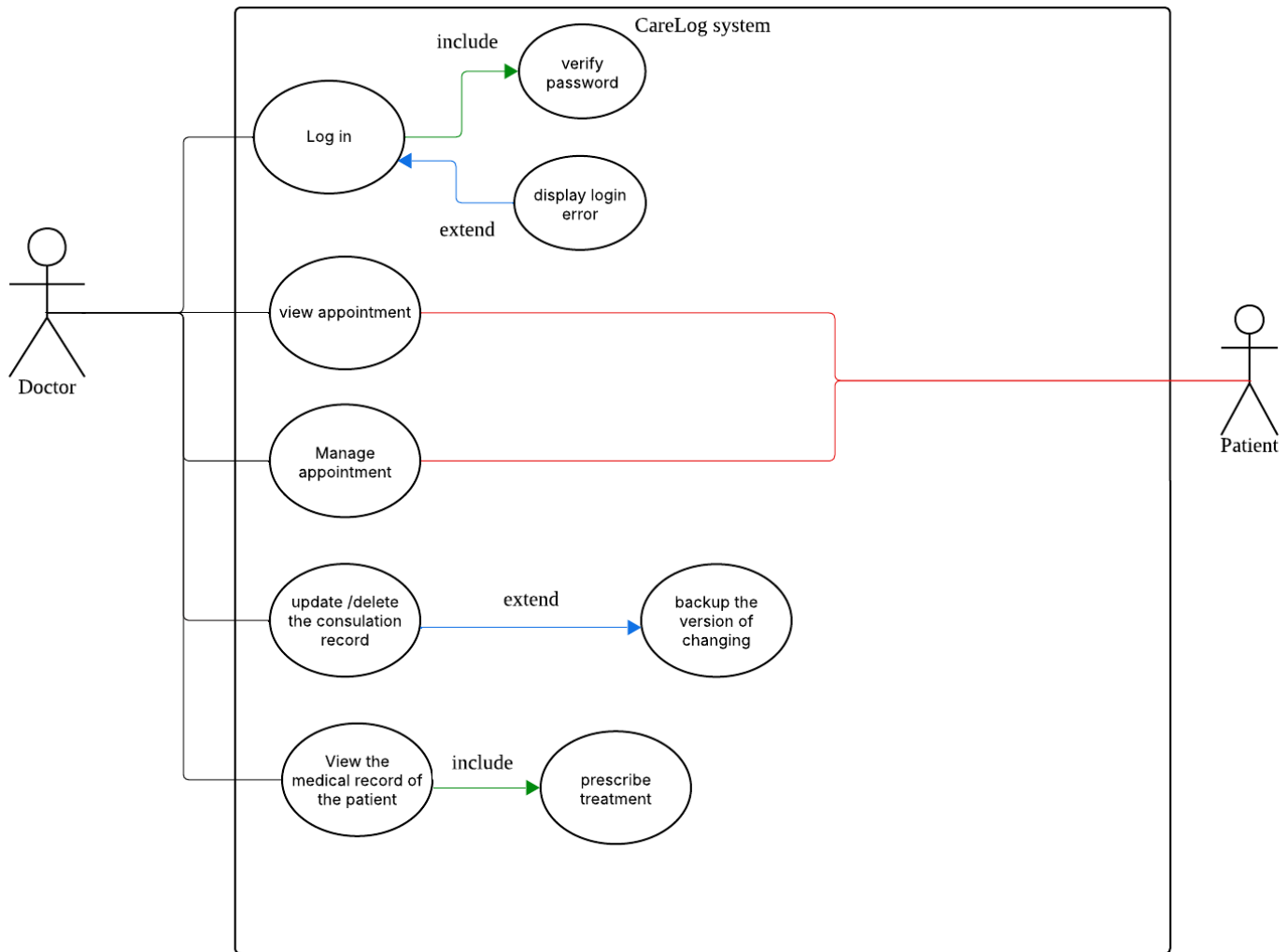
- Patient → receives direct care, medication, and monitoring.
- Medical Staff → provides observations that nurses review during care.

- Psychological Counsellor → coordinates with nurses in mental health crisis escalations. Admin → manages nurse accounts, shifts, and access controls.
- Doctor → provides treatment orders and updates.

Description: The Nurse Use Case Diagram highlights the nurse's role in administering medications, recording vitals, and supporting doctors in treatment delivery. Nurses continuously update patient charts, track tasks, and receive alerts about doctor orders or emergencies. Their role is vital for patient comfort, safety, and timely care, making them a central actor in day-to-day hospital workflows. They also serve as an intermediary between patients, doctors, and counsellors, ensuring smooth communication and coordination of treatment.

Description: The Nurse Use Case Diagram highlights the nurse's role in the log of administering medications, recording vitals, and supporting doctors in treatment delivery. Nurses continuously update patient charts, track tasks, and receive alerts about doctor orders or emergencies. Their role is vital for patient comfort, safety, and timely care, making them a central actor in day-to-day hospital workflows. They also serve as an intermediary between patients, doctors, and counsellors, ensuring smooth communication and coordination of treatment.

Doctor Use Case Diagram



Primary Actor:

- **Doctor** (main actor responsible for diagnosis, treatment, and medical decision-making).

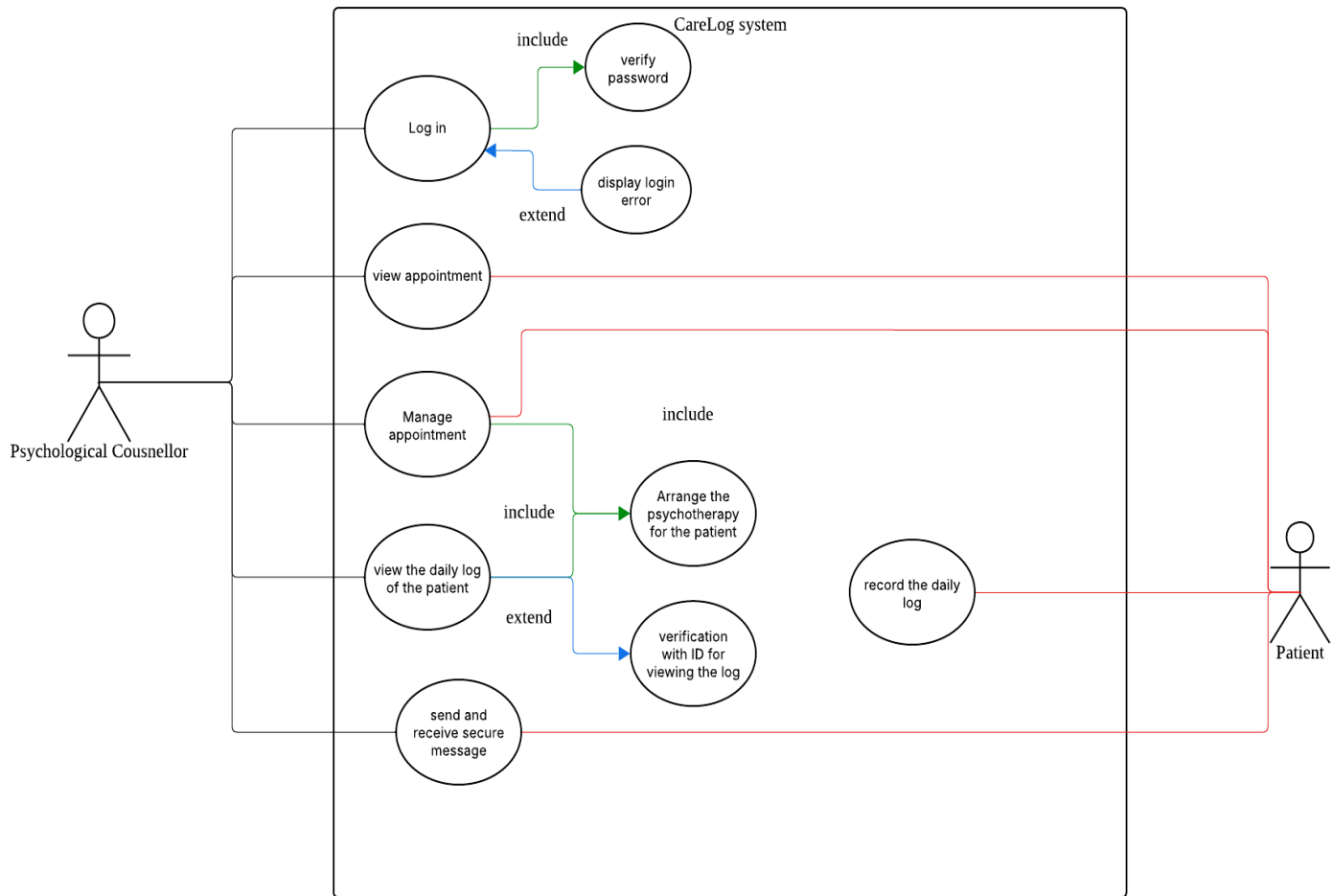
Secondary Actors:

- **Patient** → receives diagnosis, treatment, and follow-up care.
- **Nurse** → carries out treatment plans and provides progress updates.
- **Medical Staff** → provides daily notes and histories used in diagnosis.
- **Psychological Counsellor** → coordinates care where mental health impacts physical health.
- **Admin** → manages doctor profiles, schedules, and role permissions.

Description:

The Doctor Use Case Diagram highlights the role of doctors in diagnosing, prescribing, and coordinating treatment. Doctors are primary users of the system for reviewing patient history, entering encounter notes, and ordering labs or imaging. Their role directly influences patient safety and clinical outcomes. They depend on nurses and medical staff for timely updates and collaborate with counsellors for holistic patient care. Doctors also interact with admin for scheduling, credential verification, and account management.

Psychological Counsellor Use Case Diagram



Primary Actor:

- **Psychological Counsellor** (main actor supporting patient mental health and therapy).

Secondary Actors:

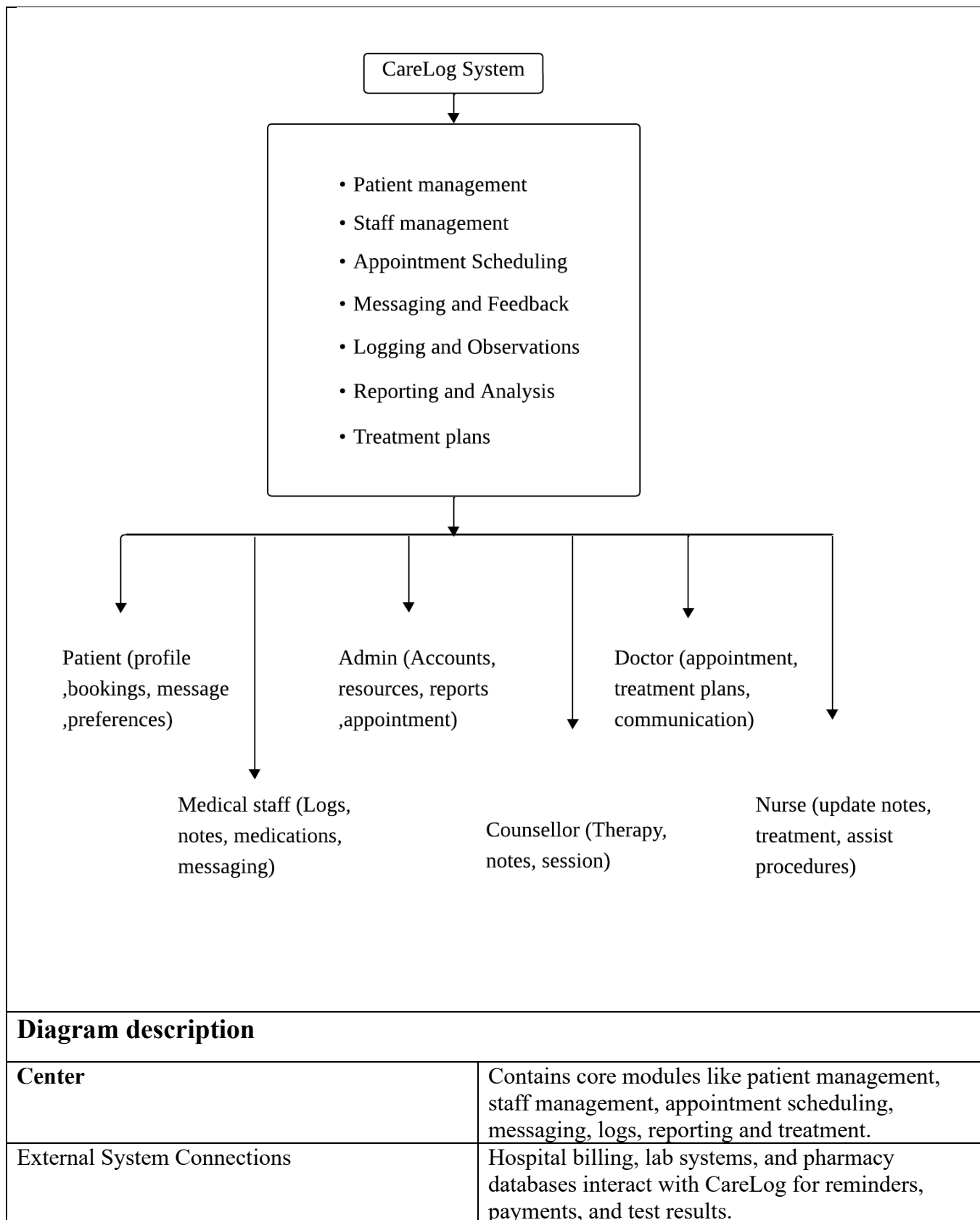
- **Patient** → attends counselling sessions, shares concerns, and follows therapy plans.
- **Doctor** → collaborates when mental health affects medical treatment.
- **Nurse** → provides updates during mental health crisis escalations.
- **Medical Staff** → supplies historical notes on patient well-being.
- **Admin** → manages counsellor profiles, schedules, and access permissions.

Description:

The Psychological Counsellor Use Case Diagram highlights the counsellor's role in providing therapy, documenting session notes, and creating safety plans. They focus on mental health progress tracking, secure communication with patients, and escalation of crisis cases to doctors or nurses. Their role is critical in ensuring that patients' psychological needs are addressed alongside medical treatment. Counsellors rely on access to patient summaries while respecting privacy and consent, ensuring confidentiality and ethical use of the system.

6. System Context Analysis

Operation Context
The CareLog system was used for patients, doctors, nurses, medical staff and psychological counsellors to interact daily in a hospital. The system is used by patients to schedule appointments, keep track of health information and communicate with staff. Physicians prescribe drugs, create treatment plans and go over patient histories. Nurses treat patients. Keep track of vital signs and update documentation. Medical staff record test results and observations to aid in diagnosis. Psychologists document therapy notes and monitor the state of mental health. Admin oversees hospital resources, scheduling, billing and user accounts, Laboratory services, pharmacy systems, billing modules and email are examples of external subsystems.
Behavioral Context
Patients log their health status, including their mood, pain, sleep patterns and medication intake. Medical persona, including nurses, then record observations and assist in the administration of treatment. Physicians make diagnoses, recommend courses of action, and keep nurses and staff informed. Counsellors use therapy notes and patient logs to track psychological states. By controlling appointments, permissions, and hospital resources, administrators supervise and plan every activity. Only authorized users can see data thanks to role-based access. While administrators maintain efficient workflows, communication loops link patients with physicians and staff.
Ecological Context
Electronic health records (EHR), laboratory information systems, pharmacy databases, and billing/payment modules are just a few of the hospital infrastructure components that the system interfaces with. By digitizing patient logs and records, CareLog lowers the amount of paper used while increasing productivity and decreasing waste in the environment. To facilitate staff mobility and patient accessibility, it also runs on a variety of devices, including PCs, tablets, and smartphones.
Sociocultural Context
By providing multilingual options and honouring cultural or religious preferences (such as dietary restrictions or nurse gender), CareLog supports a diverse patient and staff. To preserve their autonomy and dignity, patients manage their privacy settings. By allowing for individualized care and upholding social norms, the system increases trust between patients and employees. By simplifying coordination and communication, it lessens the stress of the workload for employees. It improves patient safety, comfort and involvement in decisions about their care.



Roles	
Patient	Manage accounts, logs, preferences, appointments, and communication
Medical Staff	Record observations, view logs, update treatment, manage appointments, and message patients
Admin Staff	Oversee accounts, resources, and reporting
Counsellor	Manage therapy notes, counseling appointments, and psychological support
Doctor	Arrange treatment plans(procedure) and appointment
Nurse	Keep track of patients and update notes, procedures and treatment

3. System Architecture

Patient

Entry Point

- Patient logs in via **Web/Mobile/Tablet app**.
- Authentication goes through the **Identity service** (secure login, optional MFA, lock after 3 failed attempts).

Main Interactions

1. **Profile and Consent**
 - Patient Service stores personal info and consent flags.
 - Consent Gate ensures staff can only see allowed data.
2. **Daily Logs or Surveys**
 - Patient submits well-being survey → stored in Patient Service.
 - Confirmation screen, <2s response (out-of-scope)
3. **Appointments**
 - Patient requests or confirms appointments via Appointment Service.
 - Calendar view populated from Scheduling Service.
 - Delivery confirmed, stored encrypted, audit logs track access. (out-of-scope)
4. **Notifications**
 - Review reminders/alerts in calendar
 - System remembers preferred channel.

Guarantees for patients

1. Privacy: staff only see what patient consent allows.
2. Security: data encrypted, sessions auto-logout, MFA for sensitive edits. (out-of-scope)
3. Availability: calendar/messages available ≥99.9% during hospital hours.
4. Performance: patient dashboard loads in ~2s via caching + search index. (out-of-scope)

Presentation Layer	Patient Portal (Login, surveys, messaging, appointment)
Application Layer	API Gateway, Identity, Notification
Domain, Business Layer	Patient, Clinical Notes, appointments, messaging, medications, consent gate

Data Layer	Relational Database, document store, audit log
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Medical Staff

Entry Point

- Staff logs in via MFA.
- Identity service enforces Role-Based Access Control (RBAC) to decide who can do what in the system.

Main Interactions

1. **Patient Lookup**
 - Staff queries patient → Patient Service + Search Index return summary within 2s. (out-of-scope)
 - Consent Gate filters out fields (e.g., counsellor notes hidden from general staff).
2. **Clinical Notes**
 - Doctor/Nurse enters observations in Clinical Notes Service.
 - Saved with versioning (audit trail for compliance).
 - Immediate availability to authorized colleagues (strong consistency).
3. **Medication & Orders**
 - Orders placed in Medication/Orders Service → optionally sent to Pharmacy via Integration Adapter.
 - Alerts trigger via Event Bus if urgent.
4. **Appointments**
 - Staff can view and manage patient appointments via the Appointment Service.
 - Scheduling Service prevents double-booking.
5. **Messaging**
 - Sends or receives secure messages from patients or colleagues (Messaging Service).
 - Urgent alerts escalated via Notification Service.
6. **Audit & Compliance**
 - All views/edits logged in Audit Service (who, what, when, why).
 - Staff can see “why” field is required when opening sensitive patient records.

Guarantees for staff

1. Fast access (<2s) to patient dashboards. (out-of-scope)
2. Strong consistency: no conflicting edits on clinical data.
3. Security: Multi-Factor Authentication (MFA) required, session auto-lock, all access logged.
4. Privacy enforcement: cannot bypass patient consent flags.
5. Availability: system up during shifts, with failover if integration (lab, pharmacy) is down.

Presentation Layer	Medical Staff Portal (Login with MFA, notes, orders, appointment, messaging, audit)
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Application Layer	API Gateway, identity, Notification
Domain, Business Layer	Patient, clinical notes, appointments, messaging, medications, consent gate
Data Layer	Relational Database, document store, audit log

Admin

Entry Point

- Admin logs in via **Web/Mobile/Tablet dashboard**.
- Authentication is handled by the **Identity Service** (secure login, role-based access, lock after 3 failed attempts, optional MFA).

Main Interactions

1. User & Staff Management

- Admin registers, updates, or removes staff (doctors, nurses, support staff).
- Role-based access ensures each staff member has appropriate permissions.
- Password reset and account recovery available.

2. Doctor & Resource Scheduling

- Admin assigns doctors to shifts, surgeries, and consultations.
- Scheduling Service prevents **conflicts (double-booking)**.
- Real-time updates on staff and equipment availability.

3. Patient Admissions & Room Allocation

- Admin allocates patients to available wards/rooms.
- System prevents **duplicate bed assignment**.
- Resources (equipment/rooms) updated automatically on patient discharge.

4. Billing & Financial Oversight

- Admin generates bills and oversees payments.
- Insurance claims integrated through external APIs (out-of-scope).
- Authorized staff only; full audit trail of transactions.

5. Reporting & Analytics

- Admin views dashboards on hospital performance (occupancy, billing, staff load).
- Reports exportable (PDF/Excel).
- Historical data stored for 5+ years (out-of-scope).

6. Notifications & Emergency Alerts

- Admin sends system-wide alerts to doctors/nurses.
- Emergency notifications delivered to all relevant staff within 5 seconds.
- Communication logs stored for auditing.

Guarantees for Admin

- **Security:** Role-based access control, encryption, MFA for sensitive operations.

- **Reliability:** $\geq 99.9\%$ uptime, with automatic data backup.
- **Performance:** Dashboard and reports load within $\sim 2-3s$.
- **Usability:** Simple interface requiring minimal training.
- **Scalability:** Supports increasing number of staff, patients, and departments (out-of-scope).

Presentation Layer	Admin Dashboard (User/staff management, scheduling, billing, reporting, notifications).
Application Layer	API Gateway, Identity Service, Notification Service, Scheduling Service, Billing Service, Reporting Service.
Domain / Business Layer	Staff Management, Doctor Scheduling, Patient Admissions, Room & Resource Allocation, Billing & Payments, Analytics & Reports, Emergency Alerts.
Data Layer	Relational Database (staff, patients, billing, scheduling), Document Store (records, reports), Audit Log (transactions, communication history).

Doctor architecture

1. Entry Point

- **Device Flexibility:**
 - Doctors can access the system from Web, Mobile, or Tablet App. This ensures flexibility in usage across different platforms based on the doctor's preferences.
 - Ensure that the interface adapts for different screen sizes to maintain usability across devices.
- **Authentication:**
 - Identity Service: The doctor logs in securely through the identity service, using either a single sign-on (SSO) or username/password.
 - Multi-Factor Authentication (MFA): For added security, MFA is optional but recommended, especially for sensitive actions.
 - Security Features: If there are 3 failed login attempts, the system locks the account temporarily to prevent unauthorized access, and alerts are sent to the admin for follow-up.

2. Main Interactions

- **Profile Management:**
 - The system maintains a Doctor Profile including critical details like specialty, qualifications, working hours, and contact information.
 - Doctors can update their profile easily through the application.
- **Patient Record and Encounter Notes:**
 - **Patient Data Access:** Doctors can securely view patient records, which include medical history, allergies, medications, and other relevant health data.
 - **Consent Gate:** The system ensures that only authorized data is accessible by the doctor, based on patient consent.
- **Creating and Updating Encounter Notes:**
 - Doctors can directly create, update, and store encounter notes including diagnoses, treatment plans, and prescriptions. This ensures continuity of care.
 - **Structured Notes:** The system may provide pre-defined templates for encounter notes to ensure consistency.
- **Managing Appointments:**
 - Doctors can accept, reschedule, or cancel appointments directly within the application.
 - **Calendar Integration:** The doctor's schedule is linked to a calendar where they can track appointments, availability, and patient history.
- **Secure Messaging:**
 - Doctors can send secure messages to patients and medical staff.
 - **Confidentiality:** All communication is encrypted to ensure patient privacy and comply with regulations such as HIPAA.

3. Guarantees for Doctors

- **Privacy:**
 - **Consent-based Data Access:** The system ensures that doctors only have access to the data they need based on **patient consent**. This is enforced through a **Consent Gate** mechanism.
- **Security:**
 - **Data Encryption:** All sensitive data, such as patient information and communication, is encrypted both in transit and at rest.
 - **MFA for Sensitive Actions:** For certain sensitive actions (like editing medical records or approving prescriptions), multi-factor authentication is required.
 - **Auto-Logout:** To prevent unauthorized access in case of inactivity, the system will automatically log the doctor out after a set period of time.
- **Availability:**
 - The system guarantees an uptime of $\geq 99.9\%$ during operating hours, ensuring doctors can access the system without disruption during critical hours.
- **Performance:**
 - The system is optimized for quick data retrieval, ensuring that patient records load within ~ 2 seconds, allowing doctors to make quick, informed decisions.

Presentation Layer	Doctor portal (Login, Access Patient data, Create Notes, Appointment): Doctors use the portal to manage patient records, schedule appointments and create clinical note
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Application Layer	API Gateway, Identity, Notification: Doctors authenticate via the Identity Service and receive notifications (e.g., for patient alerts, appointment).
Domain, Business Layer	Patient, Clinical Notes, Appointment, Messaging, Medications, Consent Gate: Doctors access clinical note, update appointments, manage medications and communicate with patients and staff
Data Layer	Relational database, document store, audit log Data related to doctor-patient interactions, notes and appointments are stored and tracked.

Nurse architecture:

1. Entry Point

- **Device Flexibility:**
 - Nurses access the system through Web, Mobile, or Tablet App, offering flexibility in how they interact with patient data, just like doctors and patients.
 - The user interface adapts to the screen size of the device (responsive design).
- **Authentication:**
 - Identity Service: The nurse logs in securely via the identity service, utilizing username/password.
 - Multi-Factor Authentication (MFA): MFA is optional but recommended for sensitive actions (e.g., administering medication or accessing patient records).
 - Lockout Mechanism: After three failed login attempts, the system locks the account temporarily to protect against unauthorized access.

2. Main Interactions

- **Patient Care Records:**
 - Vital Signs Logging: Nurses record vital signs (e.g., blood pressure, heart rate) in the system, ensuring accurate monitoring.
 - Medication Administration: Nurses update the system after administering medications, including dosage and time.
 - Care Updates: Nurses update the care plan, noting any changes to patient status or needs.
 - Data Integrity: Each record update is time-stamped and linked to the nurse who performed the action for auditability.
- **Care Coordination:**
 - Reviewing Doctor's Treatment Orders: Nurses receive orders from doctors and coordinate care accordingly. For example, if a doctor prescribes a medication or suggests a treatment, nurses implement the order and update patient records in real-time.
 - Treatment Progress Updates: Nurses monitor and update patient treatment progress, ensuring continuity of care between shifts.

- **Appointments and Tasks:**

- Task Management: Nurses manage their daily tasks via a to-do list or task calendar, which includes scheduled appointments, reminders, and assigned procedures.
- Task Reminders: Nurses are automatically notified of upcoming tasks (e.g., administering medications, taking vital signs, patient checkups).
- Shift Planning: The system integrates with a shift management feature, providing a clear overview of the nurse's duties for each shift.

- **Secure Messaging:**

- Communication with Doctors and Staff: Nurses can send and receive secure messages to doctors, staff, and patients, ensuring efficient and coordinated care. Messages are encrypted to protect patient confidentiality.
- Patient Communication: Nurses may also communicate with patients securely through the system, providing updates or responding to questions while ensuring data protection.

3. Guarantees for Nurses

- **Privacy:**

- Consent-based Data Access: Nurses are only authorized to access data directly related to the patients under their care, as per patient consent. They cannot access other patient data without explicit permission, ensuring patient privacy.

- **Security:**

- Encrypted Communication: All communication, whether with patients or internal staff, is encrypted to safeguard patient information and meet regulatory standards (e.g., HIPAA).
- MFA for Sensitive Actions: Nurses must use multi-factor authentication (MFA) for sensitive actions, such as medication administration or making updates to the care plan.
- Role-Based Access Control (RBAC): Nurses can only access the data and functionalities they are authorized for, based on their role and patient assignments.

- **Availability:**

- System Uptime Guarantee: The system ensures that it is available $\geq 99.9\%$ during nursing shifts, guaranteeing that nurses can always access patient data and complete their tasks without interruption.

- **Performance:**

- Quick Updates and Task Tracking: The system ensures low-latency updates for nurse tasks, with record updates and task tracking performed within ~2 seconds to enable efficient care and decision-making

Presentation Layer	Nurse portal (Login, Vital Signs, Medication Administration, Tasks): Nurses use the portal to update vital signs, track medication administration and follow treatment plans
Application Layer	API Gateway, Identity, Notification: Nurses authenticate through the Identity Service and receive task reminders or patient updates
Domain, Business Layer	Patient, Clinical Notes, Appointments, Messaging, Medication, Consent Gate:

	Nurse update patient records, vital signs and medication logs while ensuring confidentiality via the Consent Gate
Data Layer	Relational Database, Document Store, Audit Log: Nurse-related data (vital signs, medications) is stored and tracked here

Psychological counsellor architecture:

1. Entry Point

- **Device Flexibility:**
 - Counselors can access the system via Web, Mobile, or Tablet App, ensuring they can perform their duties seamlessly on different devices.
 - The application should be responsive to various screen sizes for the best user experience on mobile, tablet, or desktop.
- **Authentication:**
 - **Secure Login:** Counselors log in through the identity service using a username/password or SSO.
 - **Multi-Factor Authentication (MFA):** MFA is required for secure access, especially for actions involving sensitive patient data (e.g., reviewing therapy records, updating treatment plans).
 - **Account Lockout:** After three failed login attempts, the system locks the account temporarily to prevent unauthorized access, triggering an alert to system admins for security measures.

2. Main Interactions

- **Patient Data Access:**
 - **Therapy Records Access:** Counselors securely access patient therapy records, including past sessions and therapy notes, based on patient consent.
 - **Tracking Mental Health Progress:** Counselors can monitor patient progress, identify trends, and review prior treatments.
- **Session Notes and Treatment Plans:**
 - **Documenting Sessions:** Counselors create or update therapy session notes during or after each meeting. These notes can include patient progress, goals, and action plans.
 - **Updating Safety Plans:** Counselors update safety plans for patients, ensuring mental health care is consistent with the patient's ongoing treatment needs.
- **Appointment Management:**
 - **Session Scheduling:** Counselors can schedule, reschedule, or cancel individual or group therapy sessions through the system.
 - **Notifications & Reminders:** The system sends notifications to counselors and patients regarding upcoming appointments and schedule changes.

- **Secure Messaging:**
 - **Communication with Patients:** Counselors can send secure messages to patients regarding updates, treatment progress, or follow-ups.
 - **Staff Communication:** Counselors can also communicate securely with other healthcare providers or staff regarding patient care.

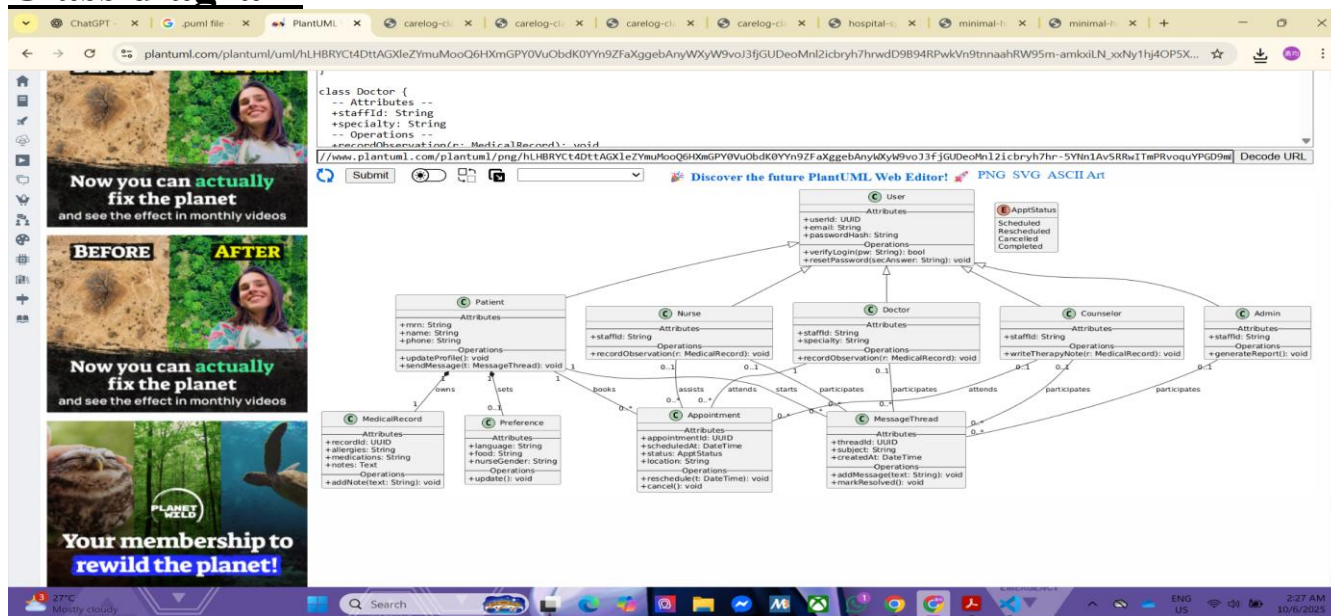
3. Guarantees for Counselors

- **Privacy:**
 - **Consent-based Data Access:** Counselors can only access patient data with explicit patient consent, ensuring privacy and adhering to confidentiality requirements. They cannot access any records without permission.
- **Security:**
 - **Data Encryption:** All therapy session notes and treatment plans are encrypted to maintain patient confidentiality and prevent unauthorized access.
 - **MFA for Sensitive Data:** Counselors are required to use multi-factor authentication for any sensitive updates or actions in the system (e.g., when creating therapy notes or updating treatment plans).
 - **Role-Based Access Control (RBAC):** Access is granted based on the counselor's role and the patient's consent, ensuring they only view data relevant to their treatment responsibilities.
- **Availability:**
 - **System Uptime:** The system guarantees $\geq 99.8\%$ uptime during therapy hours, ensuring that counselors have reliable access to the system when needed during patient sessions.
- **Performance:**
 - **Quick Record Updates:** Therapy notes and patient goals are updated and saved in real-time, with a retrieval time of ~ 2 seconds, ensuring that counselors can enter or access information efficiently during therapy sessions.

Presentation Layer	Counselor portal (Login, Therapy Notes, Appointment Scheduling, Messaging) Counselors access the portal to record therapy notes, schedule counseling sessions and communicate with patients
Application Layer	API Gateway, Identity, Notification: Counselors authenticate through Identity and receive session reminders or critical updates
Domain, Business Layer	Patient, Therapy Notes, Appointments, Messaging, Consent Gate:

	Counselors create therapy notes, track patient progress and manage appointments while respecting patient consent
Data Layer	Relational Database, Document Store, Audit Log: Therapy notes and patient communication are securely stored audited for compliance

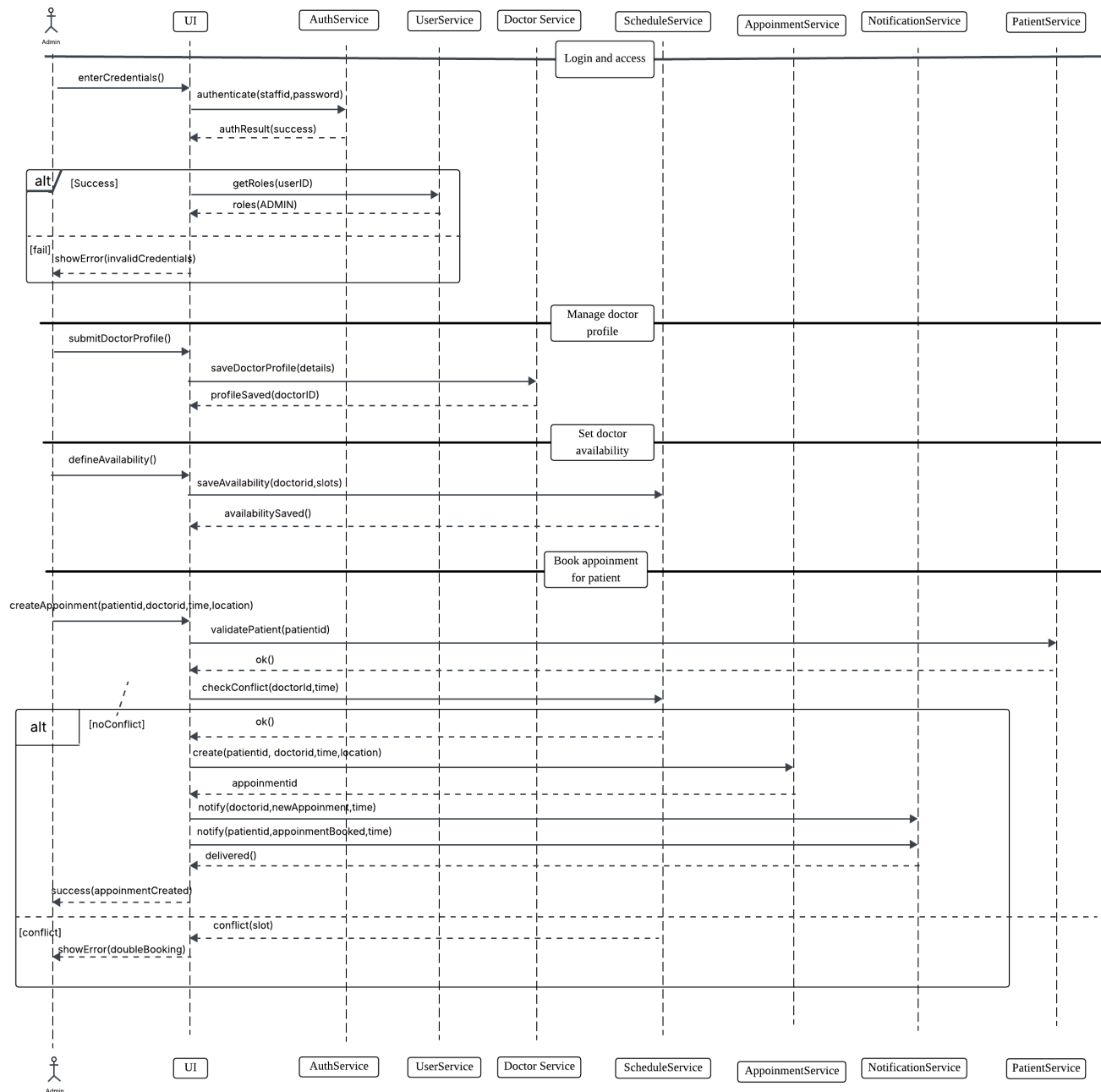
Class diagram



https://lucid.app/lucidspark/7be921c4-0840-49d8-96b5-733505754da7/edit?beaconFlowId=FEE90CAE9757B776&invitationId=inv_5b4960b9-e859-42d5-9344-e66e3d883657&page=0_0#

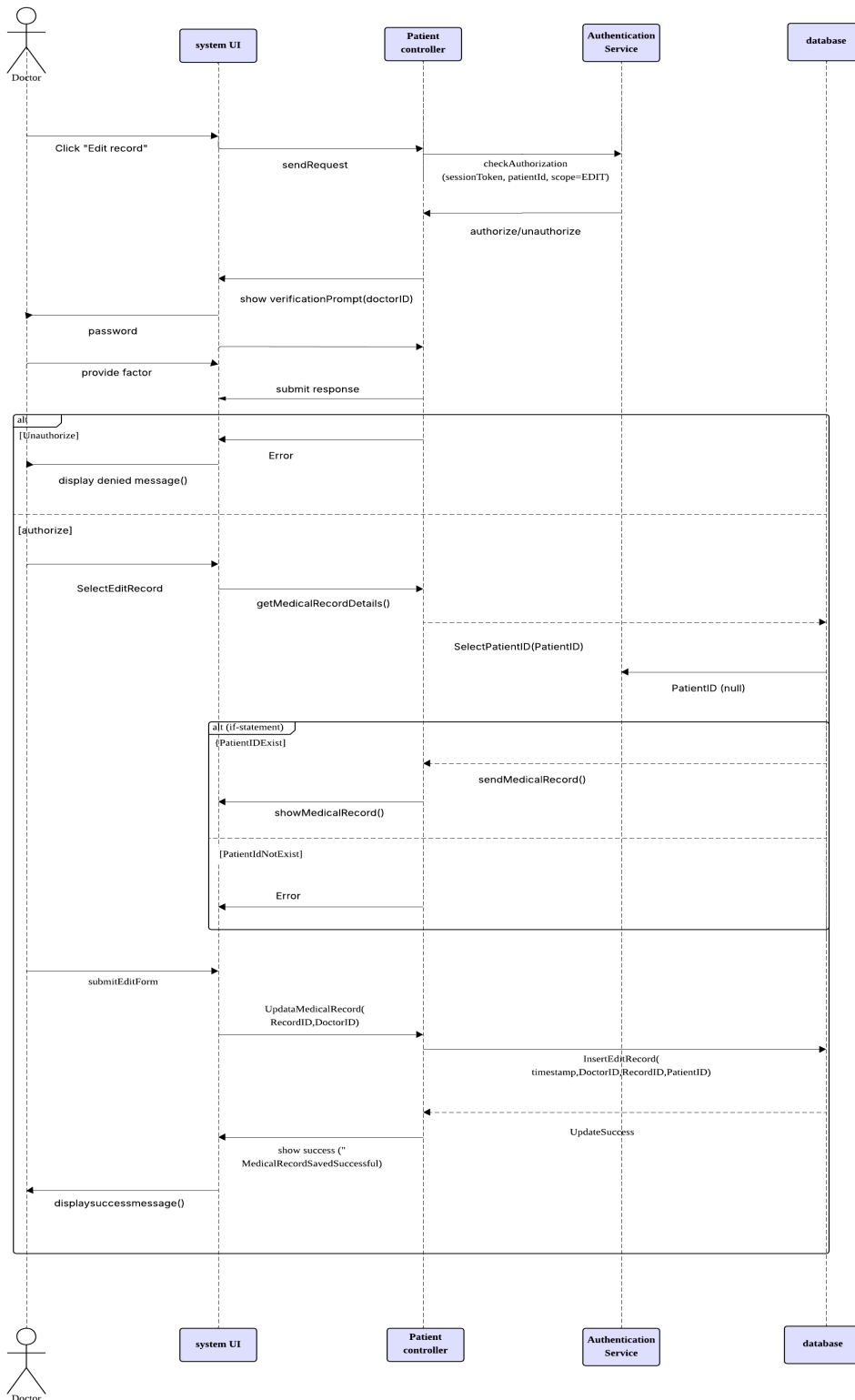
Sequence diagram

Admin



https://lucid.app/lucidspark/4be38459-5249-48fd-ad9a-7ca728eb0b63/edit?beaconFlowId=A495EDEFAD73DBF6&invitationId=inv_5bf77c30-0671-48c4-acb3-5a5c36355e09&page=0_0#

Doctor:



Doctor Logs and Edits Patient History

What the Doctor Does:

1. Views Patient History:

The doctor opens the system and searches for the patient's existing records or history.

2. Reviews and Updates Data:

If the doctor wants to add new observations, edit existing records, or update medication details, they select the log entry and edit it.

3. Submit Changes:

After editing or adding a note, the doctor submits the changes to the system.

4. Authorization Check:

The system checks if the doctor is authorized to make changes to that patient's record. If the doctor isn't authorized (e.g., they don't have access rights), the system denies access.

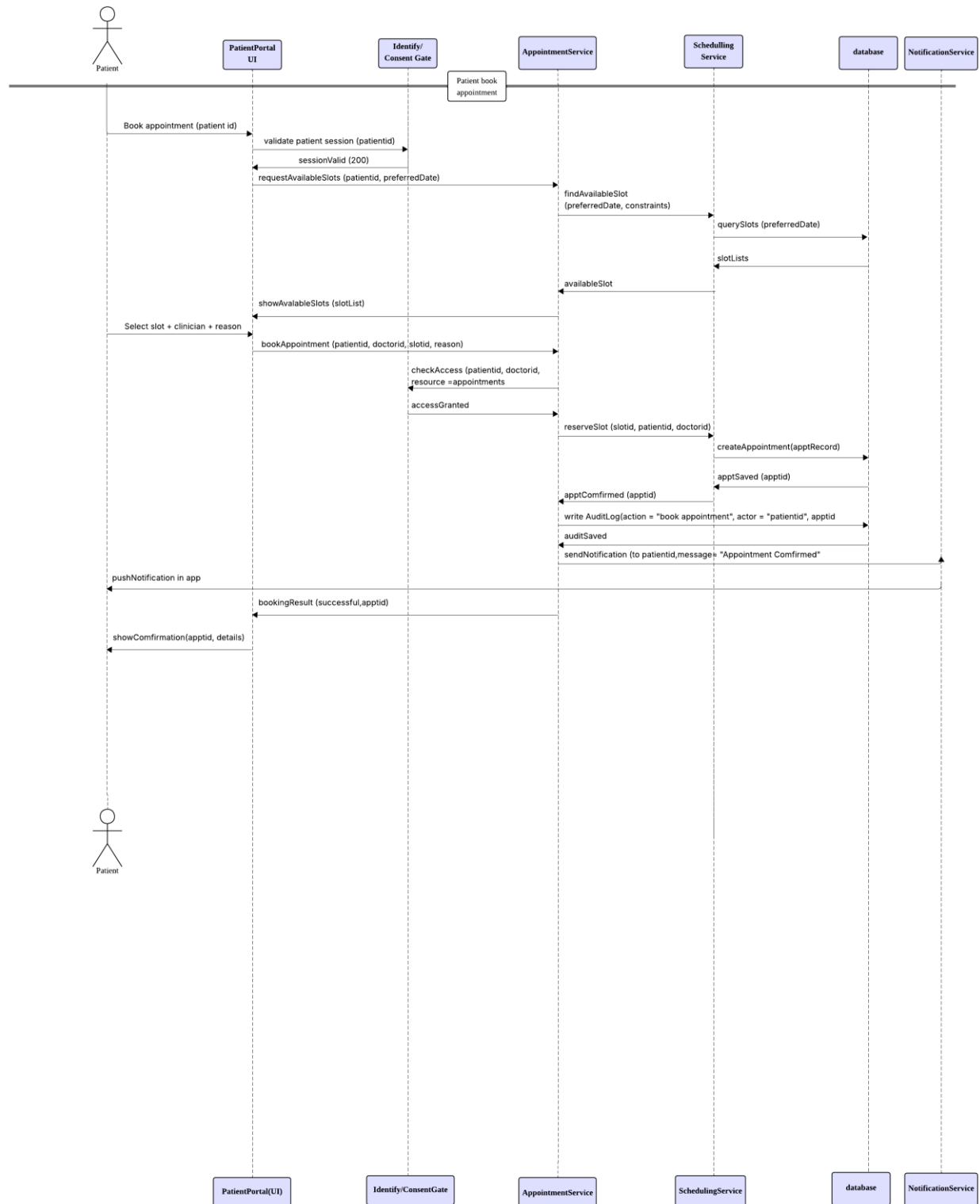
5. Validation and Data Integrity:

The system checks that any new data (like updated medication details, new observations, or edited notes) is valid and fits the expected formats.

6. Store Data and Audit:

The system records the changes and creates an audit trail to track modifications, preserving the history of changes made.

Patient



Patient Books Appointment

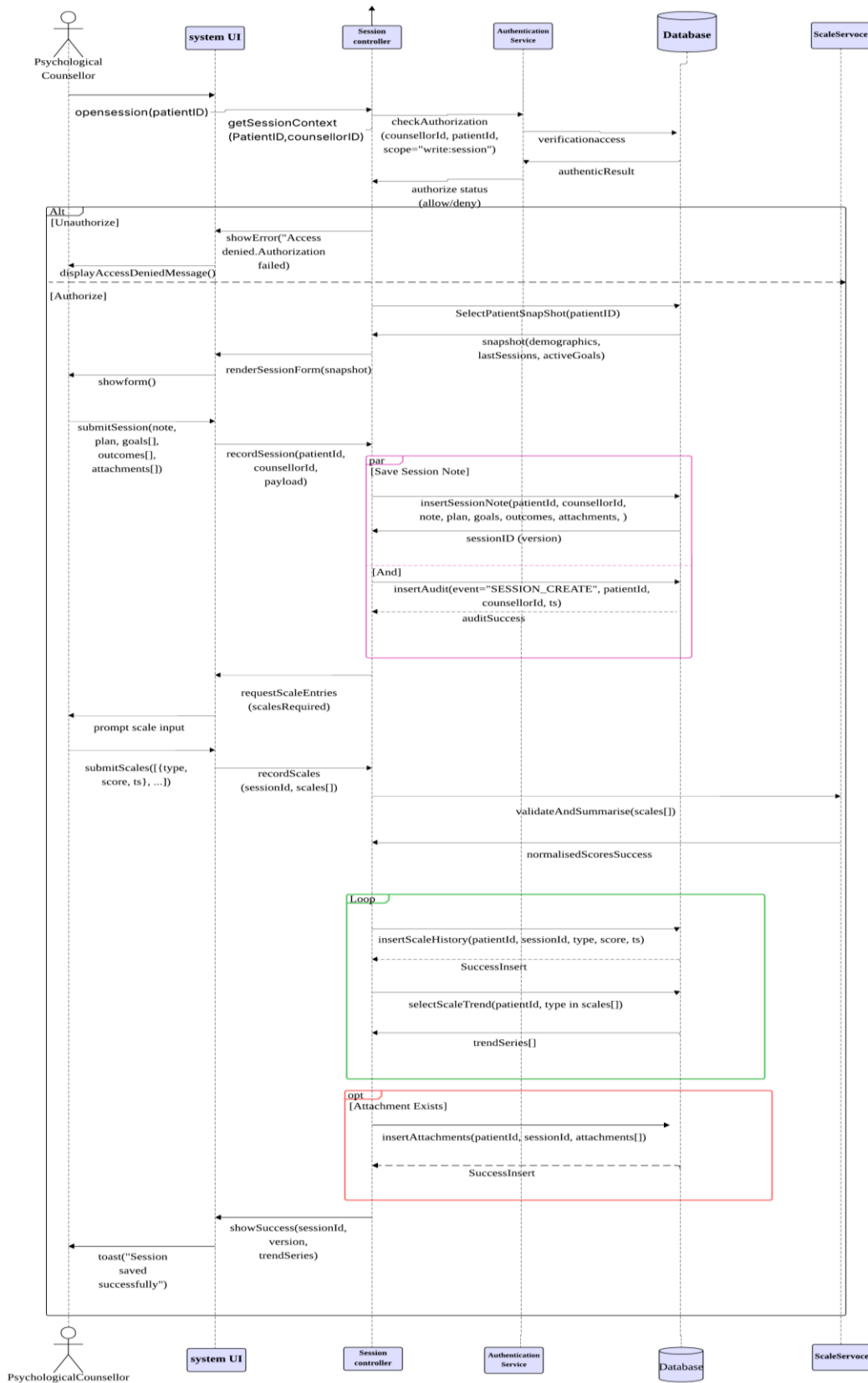
What the Patient Does:

1. Opens the Appointment Page:
The patient logs into the CareLog portal (web or mobile) and navigates to the *Book Appointment* section.
 - The page displays upcoming available dates, preferred clinicians, and appointment types (e.g., consultation, therapy, check-up).
2. Selects Preferred Slot and Doctor:
The patient chooses a preferred date, time slot, and doctor or counsellor.
 - Optional filters (e.g., clinician gender, language preference) come from the patient's saved preferences.
3. Submits Appointment Request:
The patient confirms the appointment request.
 - The portal sends the booking details to the system's Appointment Service through the API Gateway.
 - Before processing, the Consent Gate verifies that the patient has given permission for the selected clinician to access their record.

What the System Does:

4. **Session and Consent Validation:**
 - The system validates the patient's login token and checks session authenticity through the Identity Service.
 - The Consent Gate ensures the doctor or counsellor has appropriate access rights to view and book on the patient's behalf.
 - If consent or identity validation fails, the system stops the transaction and notifies the user.
5. **Check Availability and Constraints:**
 - The Appointment Service calls the Scheduling Service to verify that the selected time slot is available and that no double booking will occur.
 - The Scheduling Service queries the central database to retrieve existing bookings.
6. **Reserve and Create Appointment Record:**
 - Once the slot is confirmed, the Scheduling Service creates a new appointment entry in the database.
 - The system links the appointment to both the *Patient* and *Doctor* records.
 - The Audit Service automatically logs this action, recording the patient ID, doctor ID, date/time, and reason.
7. **Confirmation and Notification:**
 - After saving, the Appointment Service triggers the Notification Service to send a confirmation message to the patient (SMS, email, or in-app alert).
 - The notification includes date, time, clinician, and location details.
8. **System Feedback to Patient:**
 - The portal displays a success message: "Your appointment with Dr. Lim has been successfully booked for 14 Oct, 10:30 AM."
 - The appointment now appears on the patient's Upcoming Appointments list.

Psychological counsellor:



Psychological Counsellor Records Therapy Sessions

What the Psychological Counsellor Does:

1. **Opens Therapy Session Form:**

The psychological counsellor opens a therapy session form to log notes about the patient's therapy session. This could include:

Session notes (what was discussed)

- Therapy plan (treatment recommendations, next steps)
- Goals (objectives for future sessions)
- Outcomes (patient's progress or results of interventions)

2. **Input and Submit Data:**

The counsellor enters the required information and submits the form.

3. **Session Validation and Checking:**

The system validates the form, ensuring no mandatory fields are left blank and that the information entered makes sense (e.g., ensuring that no future dates are entered incorrectly).

4. **Authorization Check:**

Before the system saves the session data, it checks if the counsellor has permission to record or modify the patient's therapy records.

5. **Store and Track Progress:**

The system stores the therapy session details and creates an audit record to track when and by whom the information was recorded.

The system may also track progress against goals (e.g., progress towards therapy goals, how the patient is responding to treatment).

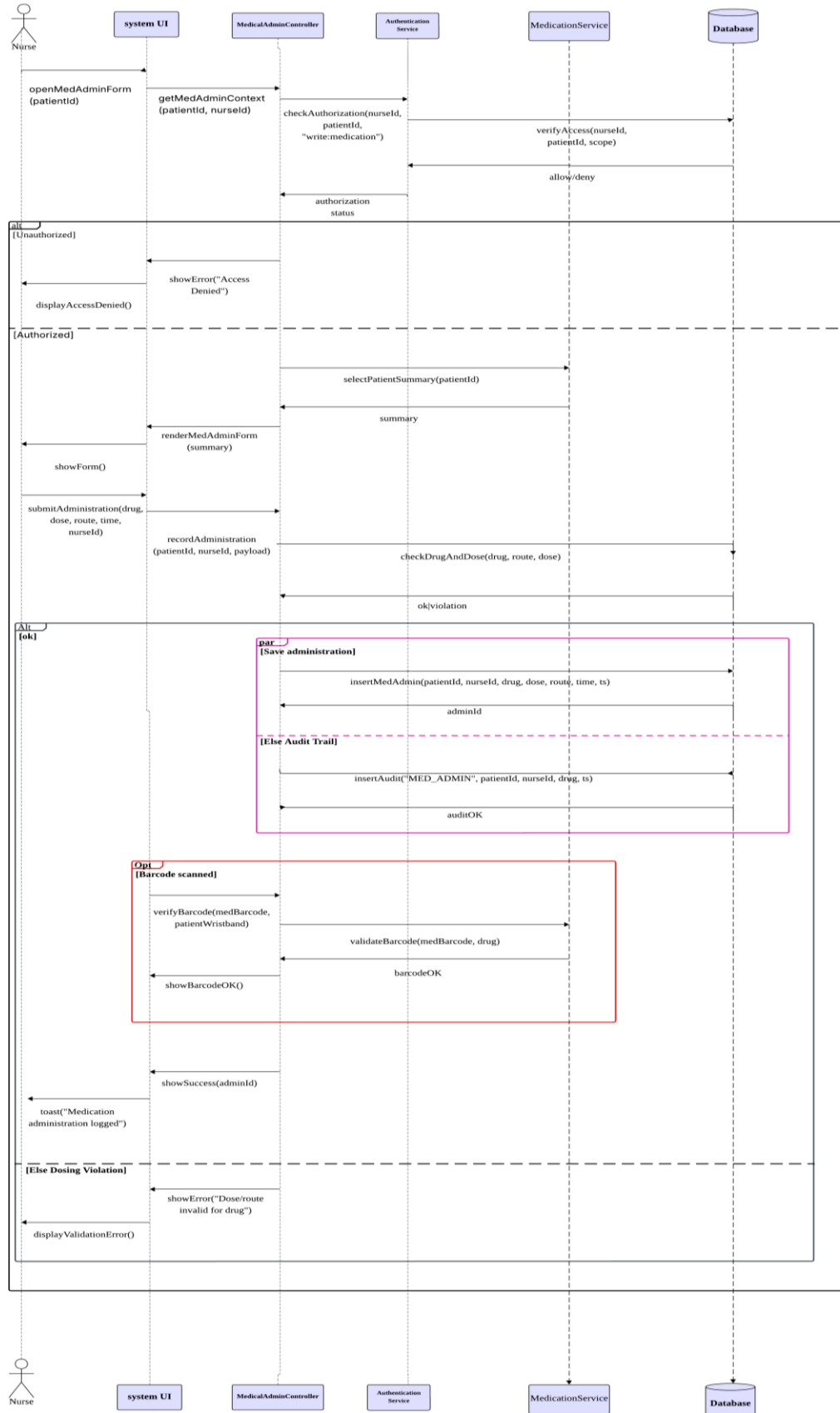
6. **Optional Scales and Assessments:**

If the counsellor uses standardized scales (e.g., mood or anxiety scales), the system validates and records those as well.

7. **Final Feedback:**

After successfully saving the session information, the system will display a success message to the counsellor, indicating that the session notes were recorded.

Nurse:



Nurse Logs Medication Administration

What the Nurse Does:

1. Opens Medication Administration Form:

The nurse opens a medication administration form to log details about the medication given to a patient. This form includes fields such as:

- Drug name (e.g., Paracetamol)
- Dose (e.g., 500 mg)
- Route of administration (e.g., orally)
- Time of administration (e.g., 10:00 AM)
- Nurse ID (to identify who administered the medication)

2. Input and Submit Medication Details:

After filling in the form with the required medication information, the nurse submits the data.

3. System Validation and Checking:

The system checks whether all required fields are filled out and if the values entered are correct (e.g., ensuring the dose is within the acceptable range for the medication).

4. Authorization Check:

Before saving the medication administration data, the system verifies if the nurse has permission to record medication details for the specific patient (checking whether the nurse is authorized to make changes to this patient's records).

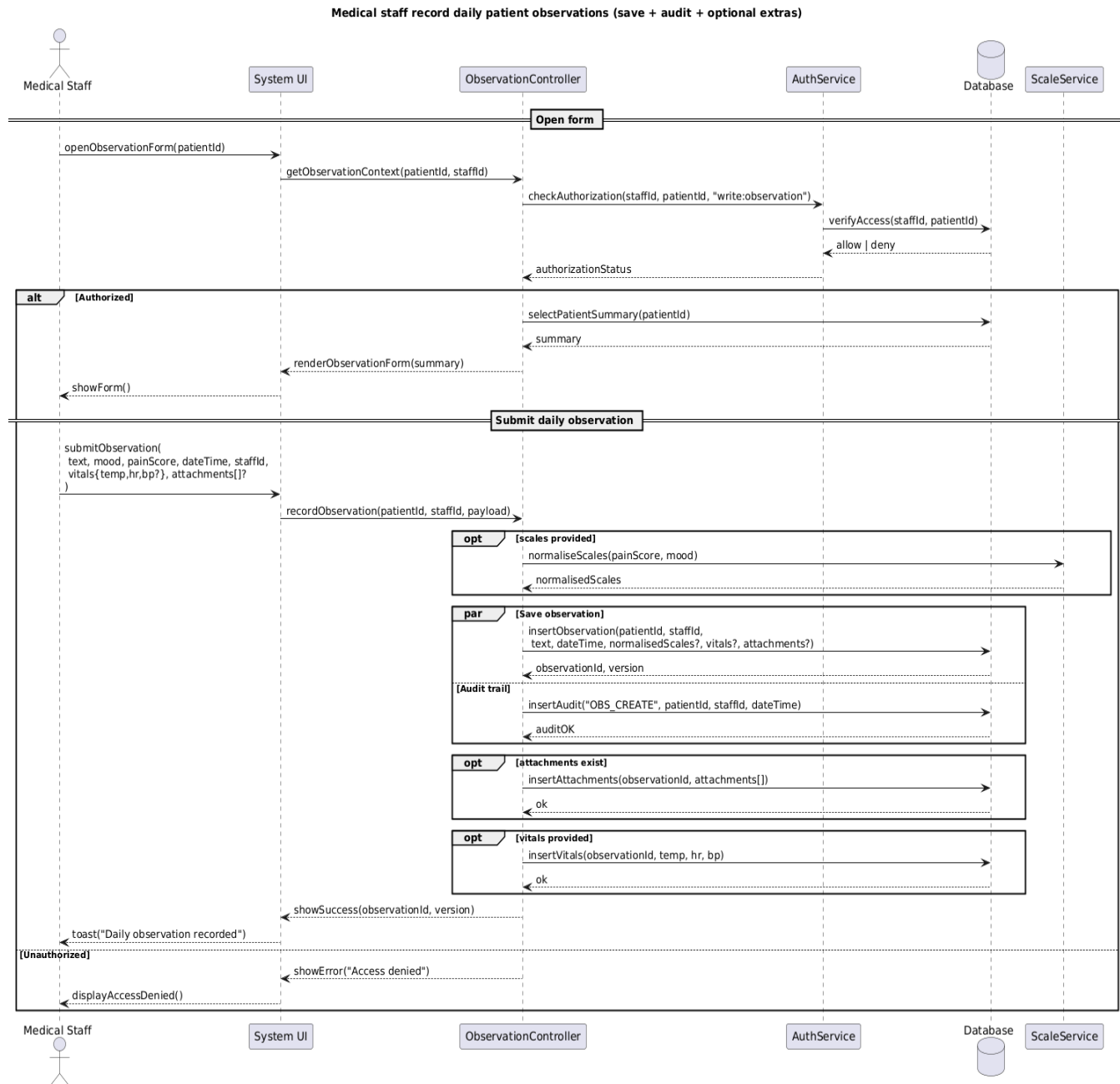
5. Store Medication Data and Audit:

If the nurse is authorized, the system saves the medication administration details in the database and creates an audit log that tracks when and by whom the medication was administered.

6. Final Feedback:

After successfully recording the medication data, the system displays a success message to the nurse, confirming that the medication administration has been logged.

Medical staff:



1. Medical Staff Logs Observations

What the Medical Staff Does:

1. Opens the Observation Form:

The medical staff (nurses, doctors) open a **form** on the system to record **daily observations** for the patient. This might include:

- Patient **mood** (happy, anxious, etc.)
 - **Pain score** (on a scale of 1 to 10)
 - **Vitals** (e.g., temperature, heart rate)
 - Any additional **notes** (e.g., changes in the patient's condition)
2. **Submit the Form:**
The staff submits the observation form, sending it to the system for validation.
 3. **Data Validation:**
The system checks if all required fields are filled and if the entered data is valid (e.g., ensuring pain scores aren't negative or that all required fields are completed).
 4. **Store Data and Audit:**
If the data is valid, the system saves the observations and creates an audit log to track who entered the information and when it was done.
 5. **Feedback to Staff:**
After submitting, the system provides feedback: either a success message or validation errors.
 - Any additional notes (e.g., changes in the patient's condition)
- **Submit the Form:**
The staff submits the observation form, sending it to the system for validation.
 - **Data Validation:**
The system checks if all required fields are filled and if the entered data is valid (e.g., ensuring pain scores aren't negative or that all required fields are completed).
 - **Store Data and Audit:**
If the data is valid, the system saves the observations and creates an audit log to track who entered the information and when it was done.
 - **Feedback to Staff:**
After submitting, the system provides feedback: either a success message or validation errors.

Usability evaluation

The primary goal of this evaluation was to assess the system's intuitiveness, efficiency, and overall user experience from the perspective of its key user groups. The evaluation was conducted using the "Think Aloud" method, where participants verbalised their thoughts while performing specific tasks.

The "Think Aloud" protocol was used. Participants were given a series of tasks and asked to say everything they were thinking, looking at, and doing as they interacted with the prototype.

- **Participants:** Two personas were recruited to represent the primary user groups:
 1. **Persona 1: Dr. Emily Chen (Hospital Care Staff)**
 - A busy junior doctor who needs to log patient notes during rounds quickly. She is tech-savvy but short on time and requires the process to be efficient and error-free.
 2. **Persona 2: Alex Rivers (Patient)**
 - A patient recovering from surgery. Alex is feeling vulnerable and has specific cultural preferences (prefers a quiet environment)
- **Tasks & Scenarios:**
 - **For Dr. Emily Chen:**
 1. **Task 1 (Login & Navigation):** "Log in to the CareLog system and navigate to the dashboard for your assigned patients."
 2. **Task 2 (Log a Clinical Entry):** "Find your patient, Alex Rivers, and log a new clinical entry. Record that his pain level is 'Moderate', his appetite is 'Poor', and he is responding well to medication."
 3. **Task 3 (Log a Personal Reflection):** "In the same log, add a personal note that Alex mentioned he prefers a quiet environment between 2-3 PM for personal prayer."
 4. **Task 4 (Review Past Logs):** "Review all past logs for Alex Rivers to get an overview of his progress."
 - **For Alex Rivers:**
 1. **Task 1 (Access & View):** "Access your personal CareLog portal from your bedside tablet. Find and review the notes that Dr. Chen left after her morning visit."
 2. **Task 2 (Add a Personal Preference):** "You've noticed the afternoon cleaning is disruptive. Add a new personal preference stating, 'Please schedule cleaning outside of 2-3 PM if possible'."

Heuristic	Application in CareLog	Strengths	Areas for Improvement
1. Visibility of System Status	The prototype uses a navigation bar and page titles.	Login confirmation and successful save messages are displayed.	Lack of loading indicators. When saving a log, the screen freezes momentarily, causing uncertainty. Dr. Chen commented, "Did it save? I'm not sure, I might click again."
2. Match between System and Real World	Uses terms like "Dashboard," "Patient Log," "Clinical Notes."	Terms like "Pain Level" and "Appetite" are familiar to clinical staff.	Jargon for Patients. Alex was confused by the term "Clinical Observations." He suggested, "Maybe call it 'Doctor's Notes' or 'My Health Updates' for me."
3. User Control and Freedom	Provides a "Back" button and "Cancel" action on forms.	Users can easily navigate away from a log without being forced to submit.	No "Undo" function. Dr. Chen accidentally deleted a log entry and had no way to recover it. A prominent "Undo" or a confirmation dialog before deletion is needed.
4. Consistency and Standards	Buttons and input fields are styled consistently.	The "Save" button is always green and located at the bottom-right of forms.	Inconsistent labelling. The button to add a new log was labelled "New Entry" in one place and "Add Log" in another, causing minor confusion.
5. Error Prevention	Dropdown menus are used for fields like "Pain Level."	Using constrained inputs prevents typos and invalid data in critical clinical fields.	No warning for unsaved changes. Alex navigated away from his "Add Preference" form without saving and lost his input. A warning dialog is essential.
6. Recognition Rather than Recall	Patient lists and past logs are visible on the dashboard.	Dr. Chen could easily see her patient list and select Alex without remembering an ID.	Hidden actions. The option to add a "Personal Reflection" was in a tab separate from "Clinical Notes." Dr. Chen initially missed it, thinking the form was only for clinical data.
7. Flexibility and Efficiency of Use	-	-	Lacks shortcuts for expert users. Dr. Chen, a frequent user, requested keyboard shortcuts (e.g., pressing 'Enter' to save) to

			speed up her workflow significantly.
8. Aesthetic and Minimalist Design	The interface is clean and uncluttered.	Both users appreciated the simple layout and lack of visual noise.	Information density could be higher. The patient dashboard felt a bit empty. Important summary info (e.g., "Last Logged: 2 hours ago") could be added.
9. Help Users Recognize, Diagnose, and Recover from Errors	Displays error messages for failed login.	Messages like "Invalid username or password" are clear.	Unhelpful error messages. When Dr. Chen submitted a form with a missing required field, the error simply said "Error." It did not specify which field was missing.
10. Help and Documentation	A "Help" section was available.	Alex found the FAQ on "How to add a preference" useful.	Documentation is reactive, not proactive. The system assumes users will seek help. Tooltips or inline hints on complex fields would be more proactive.

Strengths:

- **Intuitive Core Workflow:** The fundamental process of logging a clinical observation was straightforward for the care staff persona. The form was logically laid out.
- **Clear Visual Hierarchy:** Both users found the interface clean and easy to look at, which reduces cognitive load, especially in a high-stress hospital environment.
- **Effective Use of Constraints:** Dropdowns for clinical data (pain level, appetite) successfully prevented input errors, enhancing data quality and patient safety.

Key Areas for Improvement (Prioritized):

1. **High Priority: Data Integrity and User Confidence.** The lack of a saving indicator and an "undo" function are critical issues. They directly impact user trust in the system and can lead to data loss or duplication of work. *Fix: Implement a loading spinner and a temporary "Undo" toast notification after actions like save/delete.*
2. **High Priority: Error Prevention and Recovery.** Navigating away from a form without a warning and unhelpful error messages are major usability failures. *Fix: Implement a "You have unsaved changes" dialog and ensure all error messages are specific and actionable.*
3. **Medium Priority: Streamlining the Logging Process.** The separation of "Clinical" and "Personal" notes into tabs was not immediately obvious, hindering efficiency. For expert users, a lack of shortcuts is a missed opportunity. *Fix: Re-design the log entry form to make both sections more prominent. Plan for keyboard navigation support in a future iteration.*
4. **Medium Priority: Patient-Facing Language.** Using clinical jargon on the patient portal can be alienating. *Fix: Create a separate, simplified user interface for patients with tailored language (e.g., "Your Notes" instead of "Patient Logs").*

4. References

Picture of doctor:

<https://www.google.com/url?sa=i&url=https%3A%2F%2Fwww.vecteezy.com%2Ffree-photos%2Fdoctor&psig=AOvVaw1-sRYYIWtgELeND80eFkFh&ust=1757782093202000&source=images&cd=vfe&opi=89978449&ved=0CBUQjRxqFwoTCNjxn97W048DFQAAAAAdAAAAABAE>

Nurse picture:

<https://www.google.com/url?sa=i&url=https%3A%2F%2Fwww.healthcarescholarships.sg%2FScholarStories%2FPages%2FNursing-It%25E2%2580%2599s-Her-Calling.aspx&psig=AOvVaw0dYJdbeE1DPVUKuOQvraDl&ust=1757783822740000&source=images&cd=vfe&opi=89978449&ved=0CBUQjRxqFwoTCMjd9Zjd048DFQAAAAAdAAAAABAU>

Picture of patients:

<https://www.colourbox.com/image/portrait-of-male-patient-relaxing-in-hospital-bed-image-6241304>

Picture of medical staff:

<https://tech.co/news/microsoft-ai-tools-medical-staff>

Picture of psychological counsellor:

<https://www.visioncounselling.com.au/becoming-counselling-psychologist/>

Picture of hospital admin:https://encrypted-tbn1.gstatic.com/licensed-image?q=tbn:ANd9GcT-VWWLluyeq_32DaRkuSHHTIF52Z6Jz2NXSttjGZ7RDuj2dLzfzgPCrvhbBB6cT1DtjiTaxV2WfA0pdlMpXOczfia9sQTanC6F_znQ8voAVRM1KriWqmSm0XDavaYQTGZKzpp5vD3mrs

Case Diagram:

https://lucid.app/lucidchart/0278d055-5ce5-4dd6-bc1f-94bb0452cbe2/edit?viewport_loc=7591%2C-706%2C2966%2C1354%2C0_0&invitationId=inv_5a7165d8-6613-41ba-ad80-fddafe3e1c8f

5. Appendix